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New and Improved Probabilistic Star-Galaxy Separation Method for Rubin LSST Catalogs

Jinshuo Zhang [1], Željko Ivezić [1], Nirav Nemali,[1] Colin T. Slater,[1] Dan Taranu,[2] Robert H. Lupton,[2] and other Rubin coauthors[3]

[1] *Department of Astronomy, University of Washington, Box 351580, Seattle, WA 98195, USA*
[2] *Department of Astrophysical Sciences, Princeton University, Princeton, NJ 08544, USA*
[3] *Rubin Observatory*

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ABSTRACT

The morphological classification of resolved and unresolved sources, commonly referred to as "star-galaxy separation", is a fundamental step in the analysis of wide-field imaging surveys such as Rubin Observatory's Legacy Survey of Space and Time (LSST). Star-galaxy separation becomes increasingly challenging at the faint magnitudes probed by LSST because galaxy counts rise rapidly with magnitude while their angular sizes become progressively smaller. LSST photometric catalogs provide several classifiers, including a per-band binary classifier `extendedness` based on the threshold difference between PSF, $m_{\rm PSF}$, and CModel, $m_{\rm CModel}$, magnitudes that was pioneered by SDSS, and a novel method `griz_model_extendedness` that uses best-fit Sersic model parameters and also corrects for noise and size effects. All provided classifiers suffer from several theoretical and practical shortcomings. In this work, we implement the Bayesian star-galaxy classification method of Slater, Ivezić & Lupton (2020, SIL2020), which overcomes these shortcomings. In each band, we model the distribution of sources in the two-dimensional space defined by $\Delta m = m_{\rm PSF} - m_{\rm Sersic}$ and magnitude $m_{\rm Sersic}$. Importantly, CModel magnitudes are replaced by Sérsic magnitudes because the latter result in superior performance. At a fixed magnitude, the distribution is modeled as a mixture of unresolved and resolved components which automatically accounts for the varying star-to-galaxy number ratio and allows us to compute the posterior probability that a source is resolved, as well as sample completeness and contamination for arbitrary selection thresholds. This approach also allows for a statistically sound combination of morphological information from multiple bands. We apply this method to the Rubin Data Preview 2 object catalog and validate the resulting classifications using external HST labels in the COSMOS field with 5σ detection limit for point sources in the r band of 26.4. Using these labels as the ground truth, we quantitatively compare the performance of the Bayesian classifier with that of the default Rubin `extendedness` parameter and find that the former consistently outperforms the latter, with the improvement increasing as additional bands are incorporated. Approximately, the Bayesian classifier achieves the same classification performance for stellar subsamples at SNR limits *more than a magnitude fainter* than the CModel-based `extendedness` classifier, and approaches idealized statistical limit discussed by SIL2020. In addition, it allows users to control sample completeness and/or contamination as a function of magnitude. Depending on scientific applications, this morphological classification can be further improved by incorporating complementary color-based classifiers.

Keywords: Galaxies (573) — Cosmology (343) — Stellar astronomy (1583) — Solar physics (1476)

Corresponding author: Željko Ivezić
Email: ivezic@uw.edu

1. INTRODUCTION

The morphological classification of resolved and unresolved sources[4], commonly referred to as "star-galaxy separation", is a fundamental step in the analysis of wide-field imaging surveys such as the Rubin's Legacy Survey of Space and Time (LSST). At bright magnitudes this problem is relatively simple, since stars outnumber galaxies, and galaxies that exist are obviously extended on the sky. Star-galaxy separation becomes increasingly harder at faint magnitudes, such as those probed by Rubin's Legacy Survey of Space and Time (LSST; for details see Ž. Ivezić et al. 2019a and references therein) because galaxy counts rapidly increase with magnitude, while their angular sizes become smaller. The selection of clean faint stellar samples with high completeness is especially challenging at high galactic latitudes.

A variety of methods have been proposed in the literature for morphological classification of sources in images. C. T. Slater et al. (2020) (hereafter SIL2020) discussed the statistical foundations of morphological star–galaxy separation and showed that many of the star-galaxy separation metrics in common use today, e.g., by Sloan Digital Sky Survey (SDSS) *photo* pipeline (R. H. Lupton et al. 2002), or SExtractor pipeline (E. Bertin & S. Arnouts 1996), are closely related both to each other, and to the Bayesian model odds ratio, first derived by W. L. Sebok (1979). SIL2020 quantified the scaling of various algorithms with the source noise properties and showed that the Bayesian model odds ratio achieves the best peformance. They also showed how to probabilistically combine multiple measurements, either of the same type (e.g., subsequent exposures), or differing types (e.g., multiple bandpasses), or differing methodologies (e.g., morphological and color-based classification).

Following SDSS, LSST photometric catalogs use a binary classifier `extendedness` in each band, based on the threshold difference between PSF, $m_{\rm PSF}$, and CModel, $m_{\rm CModel}$, magnitudes (described in detail in the next section), $\Delta m = m_{\rm PSF} - m_{\rm CModel}$. For unresolved sources, this Δm is vanishing within noise, with `extendedness=0`, while for resolved objects, selected by $\Delta m > 0.016$ mag, `extendedness=1`. Resolved objects have positive Δm because their PSF flux is smaller than CModel flux due to a narrow weighting PSF profile. For an illustration of Δm distribution in Rubin's Data Preview 1 catalogs, see Figure 28 in Vera C. Rubin Observatory Team et al. (2026) or Figure 3 further below). We also discuss a novel method introduced by Rubin (R. Observatory Science Pipelines Developers et al. 2026), named `griz_model_extendedness`, that uses best-fit Sersic model parameters and additionally corrects Δm information for noise and size effects (more details are available in the next section).

Although `extendedness` was successfuly used in thousands of papers based on SDSS catalogs, this approach is not optimal for LSST. As discussed in SIL2020, when the star-galaxy separation threshold is optimized at the faint, low signal-to-noise ratio (SNR) end of the data, the ability to recognize barely resolved objects at the bright end (such as close binary stars) is not fully exploited. Furthermore, the binary nature of `extendedness` parameter prevents optimal combination of information from different bands, as well as from other classification methods (e.g., based on colors, or other datasets). SIL2020 showed how to turn Δm into a probability distribution, which then allows proper probabilistic combination of multiple bands: with appropriately chosen priors for star-to-galaxy counts ratio, the posterior probability that a source is unresolved (or resolved) can be computed in a statistically sound way using Bayesian framwork.

The main goals of this paper are to implement the steps towards Bayesian star-galaxy separation outlined in SIL2020 using Rubin's Data Preview 2 data and quantify its performance. We introduce our dataset in §2, and in §3 describe star-galaxy separation methodology, including quantification of its performance using HST-based labels. We discuss our results in §4.

2. DATASETS

Here we describe two datasets used in our analysis: Rubin Data Preview 2 catalogs and HST data with star-galaxy labels used for testing methods discussed in §3.

[4] Note that quasars and galaxies can be unresolved and even some stars can be resolved, such as Asymptotic Giant Branch stars. Here we use interchangeably star/galaxy and unresolved/resolved nomenclature when referring to morphological classification.

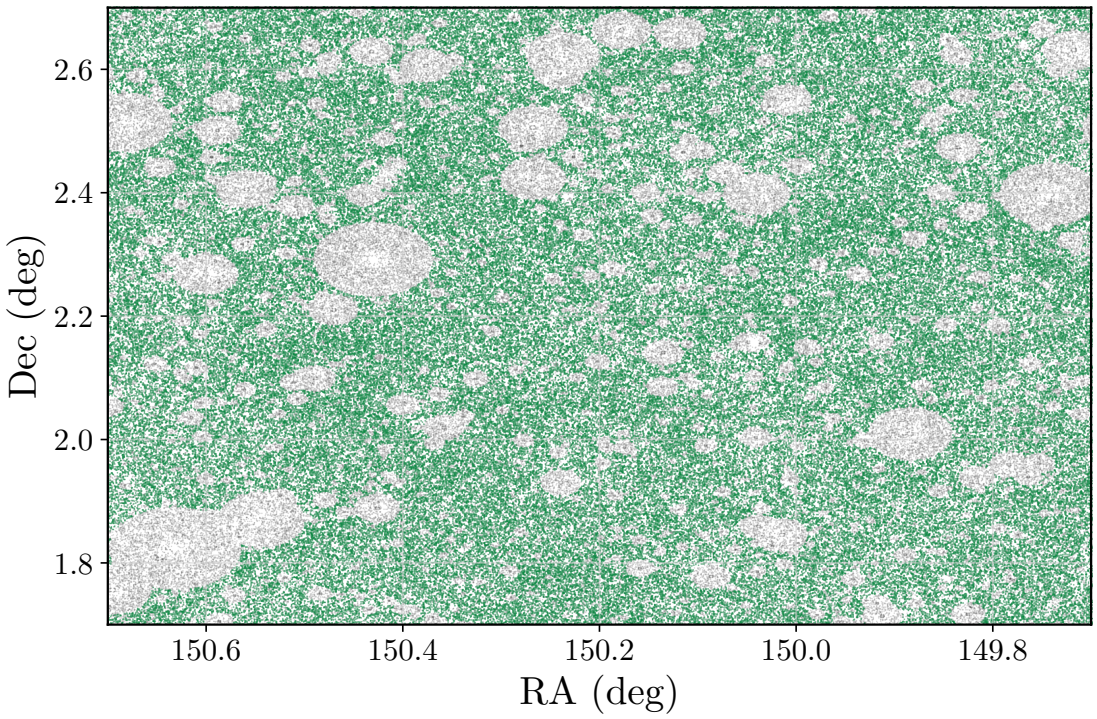


Figure 1. The central DP2-COSMOS region used for the dataset overview. Gray points show Rubin DP2 objects and green points show the subset matched to the HST COSMOS2020/FARMER sample using a 0.5 arcsec matching radius. The white regions in the centers of gray areas correspond to areas where no objects are listed in either catalog.

2.1. *Rubin Data Preview 2*

All data collected with the LSST Camera (Vera C. Rubin Observatory Team 2026) during the commissioning of the Rubin Observatory will be released as Data Preview 2[5] (hereafter DP2). Coadded deep images and resulting catalogs have been released in July 2026 as Early Data Preview 2. These coadded images are based on about 24,000 single-visit images and cover about 3,000 sq. deg. of sky.

In our analysis we focus on the COSMOS field because it is both already well observed by Rubin (to only about 1 magnitude shallower limit than anticipated for 10-year LSST), and space-based HST COSMOS2020/FARMER star and galaxy labels are available (J. R. Weaver et al. 2022). As a control sample for testing code robustness, we also used Rubin data for the Extended Chandra Deep Field South (ECDFS). The Rubin 5σ depth for point sources in the COSMOS field, estimated as the median magnitude of point sources with photometric errors of $\sim$0.2 mag, is (25.66, 26.66, 26.36, 25.91, 24.94, 23.59) in the *ugrizy* bands, respectively.

2.2. *HST Star-galaxy Classification Labels in the COSMOS Field*

For the COSMOS validation sample, we select Rubin DP2 sources in the rectangular region $149.7 < \mathrm{RA} < 150.7$ and $1.7 < \mathrm{Dec} < 2.7$, with $16 \leq r_{\mathrm{CModel}} < 26.5$. Objects listed in the COSMOS2020/FARMER reference catalog are matched to DP2 sources using a 0.5 arcsec matching radius. Within the selected region and magnitude ranges, the DP2 sample contains 269,606 objects, of which 193,586 are matched to the COSMOS2020/FARMER objects and 76,020 are unmatched (a matching fraction of 72%). This somewhat low matching fraction is due to large areas around bright stars that are masked in the COSMOS2020/FARMER catalog (see Figure 1).

[5] See https://dp2.lsst.io/

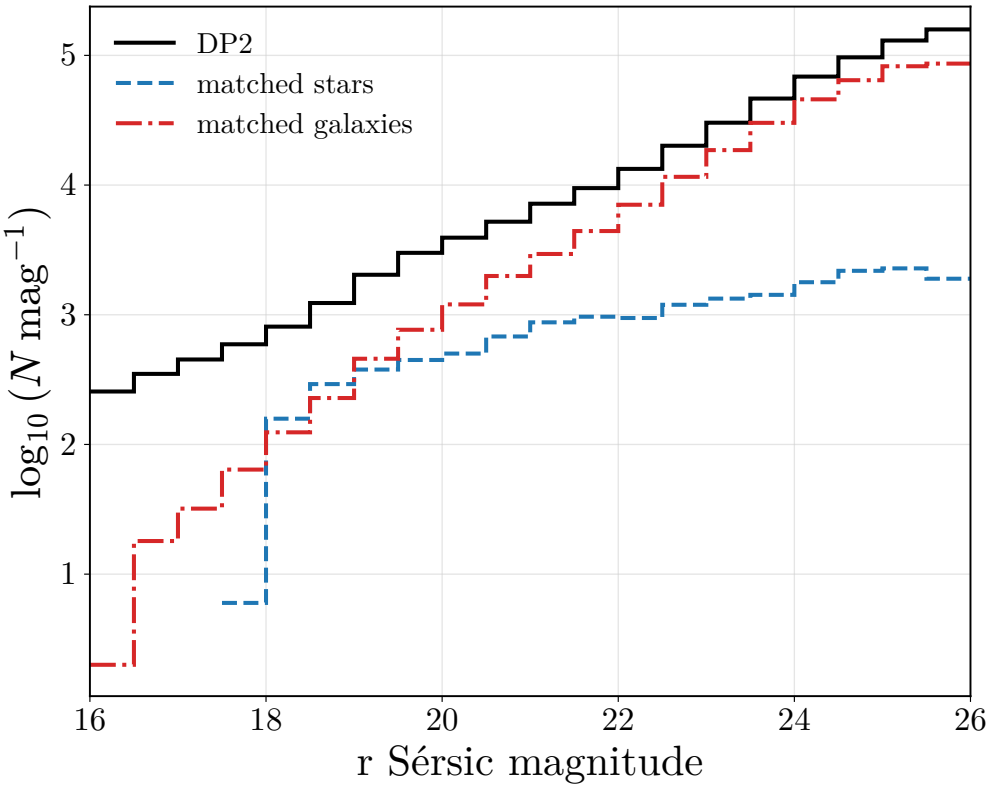


Figure 2. Magnitude distribution of the DP2-COSMOS sample. The black curve shows all selected DP2 sources, while the blue and red curves show matched COSMOS2020/FARMER sources classified as stars and galaxies, respectively. Counts are shown per magnitude, using 0.5 mag wide bins. Note that galaxies outnumber stars at the faint end by more than an order of magnitude.

The matched validation sample contains 8,788 stars and 184,798 galaxies, classified using HST's `truth_binary` = 1 for stars and `truth_binary` = 0 for galaxies. Their number counts as a function of magnitude are shown in Figure 2. The low fraction stars is due to very faint sample limit $r \sim 26$) and high Galactic latitude ($b \sim 50°$).

2.3. *The color-magnitude distributions of sources classified using HST labels*

The external COSMOS2020/FARMER labels provide the reference classification, considered here as ground truth, for examining the distribution of stars and galaxies in various diagrams and for measuring the performance of star-galaxy separation methods. Nevertheless, it is likely that some very faint sources classified by HST as stars might be unresolved galaxies even at HST's resolution. Therefore, the HST labels will help us assess whether a source should be resolved or unresolved in Rubin images, but do not necessarily provide definitive information about astrophysical nature of a source (e.g., to separate unresolved stars and galaxies, one would need spectra or broad-band photometry over a much wider weavelength range, say, in the near-infrared K band). Furthermore, no classification is perfect and thus we first examine the distribution of HST-classified "stars" and "galaxies" in various informative color-color diagrams constructed with Rubin photometry.

The top left panel in Figure 3 shows the distribution of stars and galaxies classified using HST labels in the $\Delta m = m_{\rm PSF} - m_{\rm Sersic}$ vs. $m_{\rm Sersic}$ diagram (justification for replacing CModel magnitude by Sérsic magnitudes in Δm is discussed in §3). As we describe in detail in the next Section, modeling this distribution is a key ingredient of the Bayesian star-galaxy separation method presented here.

The top right panel in Figure 3 shows that faint galaxies ($r > 24$), in addition to outnumbering stars at faint magnitudes, have on average bluer $g - i$ colors (based on CModel magnitudes and corrected for ISM dust reddening using values provided in DP2 catalogs) than stars. The clearly different distributions of stars

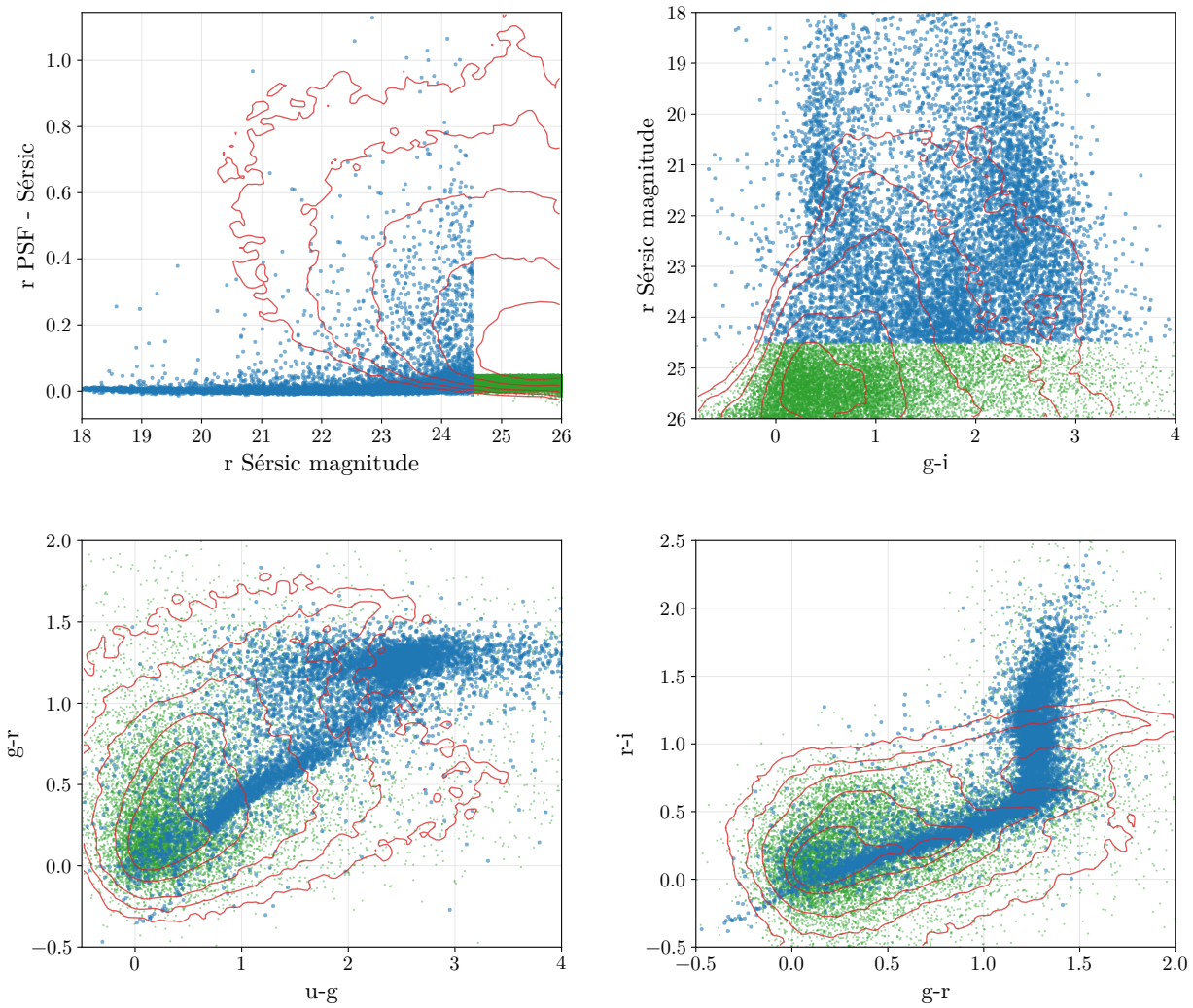


Figure 3. The distribution of stars with r-band Sérsic magnitude < 24.5 (blue dots) and galaxies (red contours), classified using HST labels, in various informative diagrams. Green points show Rubin DP2 sources with r-band Sérsic magnitude $24.5 < r < 26$ and $m_{\rm PSF} - m_{\rm Sérsic} < 0.05$. The CModel-magnitude based colors are corrected for the ISM dust extinction, while the r-band Sérsic magnitude is not. The top left panel shows the $\Delta m = m_{\rm PSF} - m_{\rm Sérsic}$ vs. $m_{\rm Sérsic}$ diagram, where stars form a narrow sequence centered on $\Delta m \approx 0$ and galaxies generally have $\Delta m > 0$. The top right panel shows the r vs. $g-i$ color magnitude diagram, and the two bottom panels show the $u-g$ vs. $g-r$ and $g-r$ vs. $r-i$ diagrams. The main stellar locus, dominated by main sequence stars and red giants, is clearly visible with its blue end redwards from $u-g \sim 0.6$; unresolved sources with $u-g < 0.6$ are expected to be a mixture of low-redshift quasars, white dwarfs and unresolved binaries. In the region with $24 < r < 24.5$ and $0.0 < g-i < 1.0$, galaxies outnumber stars by a factor of 82.

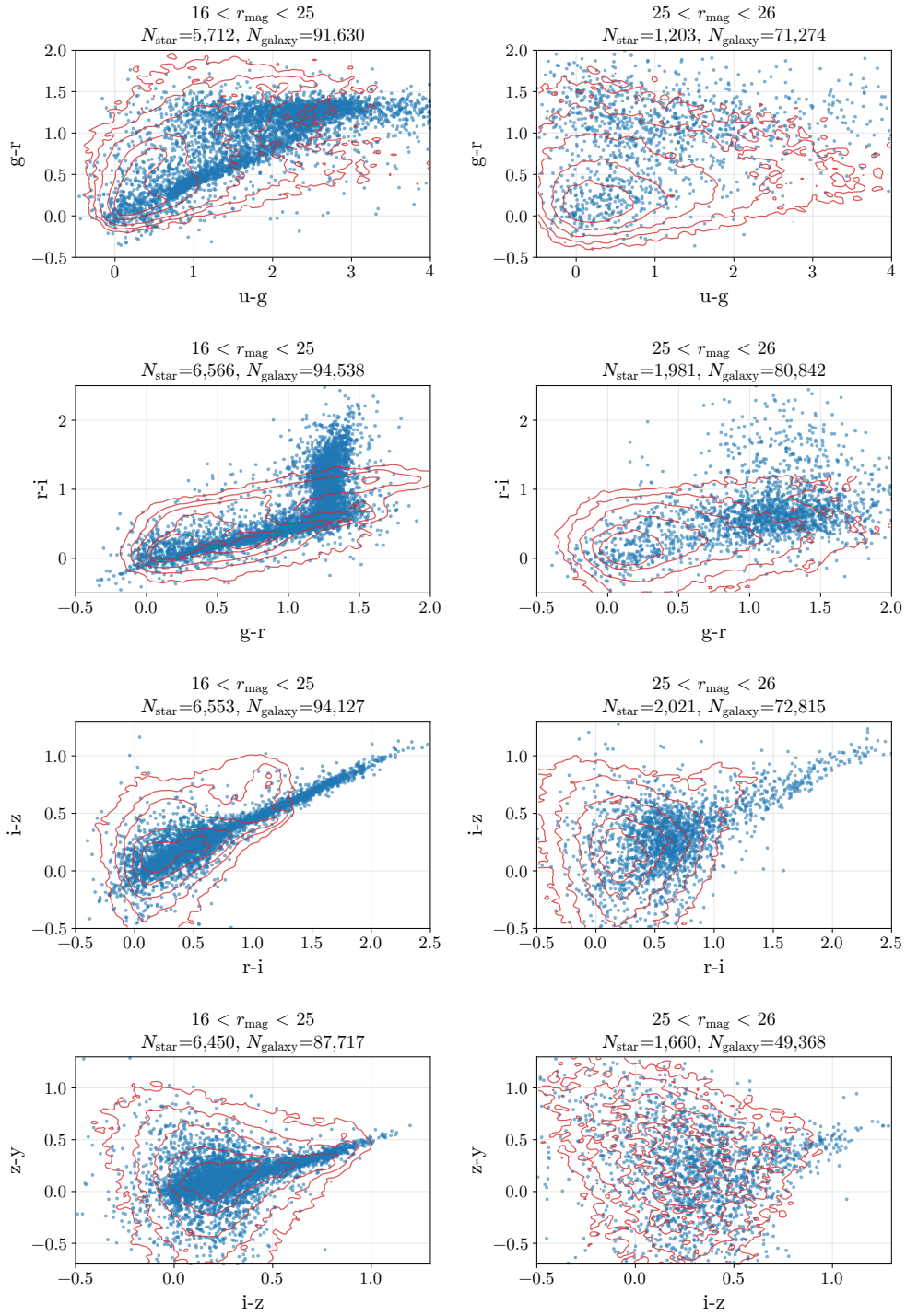


Figure 4. Similar to the bottom two panels in Figure 3, except that here all principal color-color diagrams are shown and the data are split by the r-band Sérsic magnitude to show deterioration of photometry close to the faint magnitude limit (left: $16 < r_{\rm mag} < 25$; right: $25 < r_{\rm mag} < 26$). The 5σ depth for point sources in the r band for this dataset is 26.36.

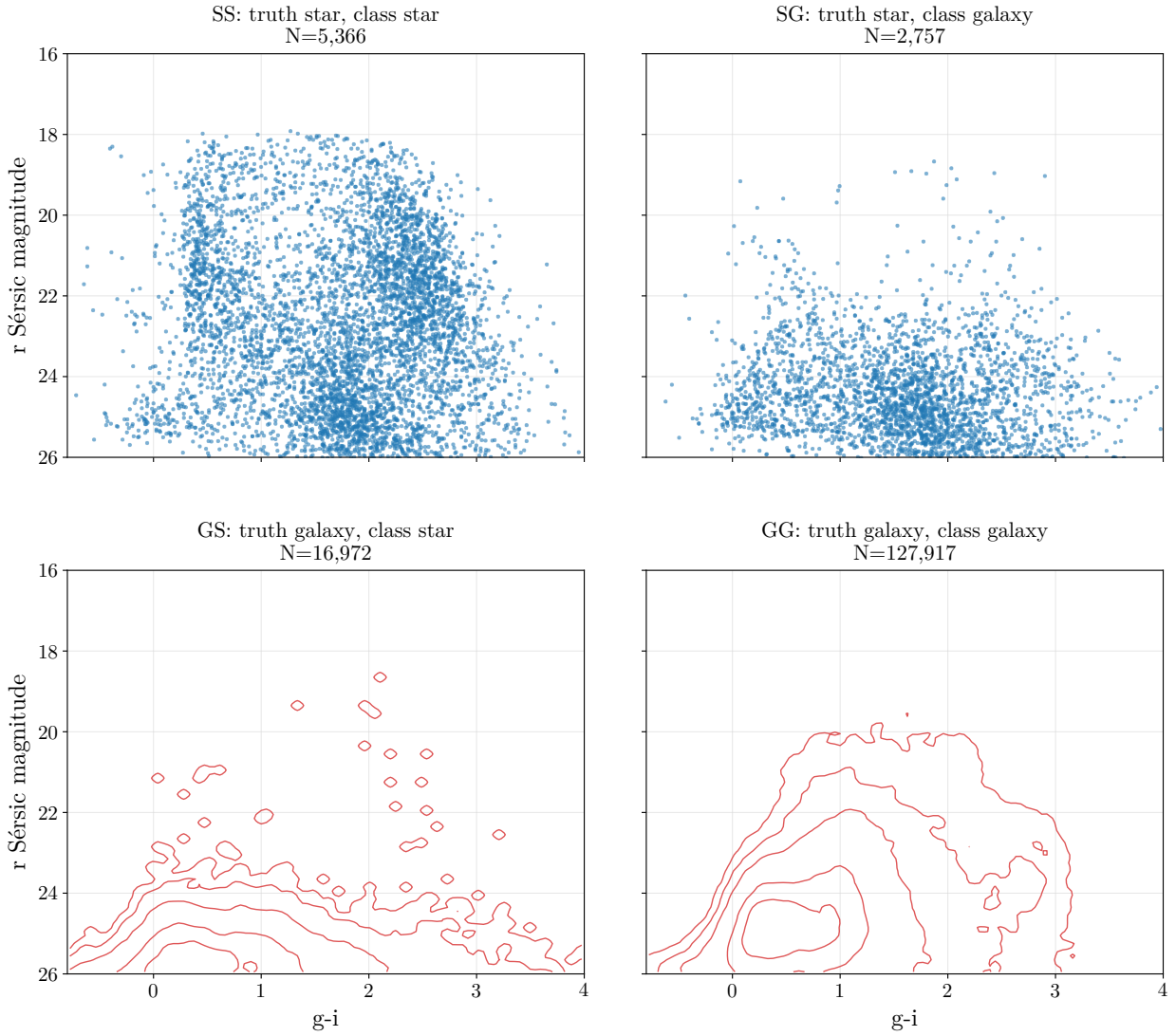


Figure 5. Performance of Rubin's default single-band binary `extendedness` classifier in the r band relative to HST ground-truth labels. The top left and bottom right panels show correctly classified unresolved and resolved sources in the r vs. $g-i$ color-magnitude diagram, with counts shown in the title above each panel. The bottom left and the top right panels show misclassified sources. Note that the stellar sample selected by `extendedness`=0 is severely contaminated by galaxies for $r_{\rm Sérsic} > 24$ (bottom left panel). In the region with $24 < r < 24.5$ and $0.0 < g-i < 1.0$, misclassified galaxies (bottom left panel) outnumber selected stars (top left panel) by a factor of 5.5.

and galaxies in color-color diagrams shown in the bottom two panels in Figure 3 demonstrate that colors can be used to enhance morphological star-galaxy separation. Nevertheless, Figure 4 and the top left panel in Figure 3 show that increased photometric noise close to the faint magnitude limit will cause deteriorated performance for both morphological and color-based star-galaxy separation methods. Extending the start of this deterioration to a fainter magnitude limit is one of the primary goals of this work. It is notable that the top right panel in Figure 4 implies that most unresolved faint blue sources have $u-g$ color more typical of low-redshift quasars ($u-g < 0.6$) than of main sequence stars (for a detailed discussion of the blue star counts see L. Palaversa et al. 2026).

2.4. *The color-magnitude distributions of sources classified using* `extendedness` *parameter*

Difficulties with star-galaxy separation encountered at the faint end are illustrated in Figure 5, which compares subsamples classified using HST ground-truth labels and the default Rubin `extendedness` binary

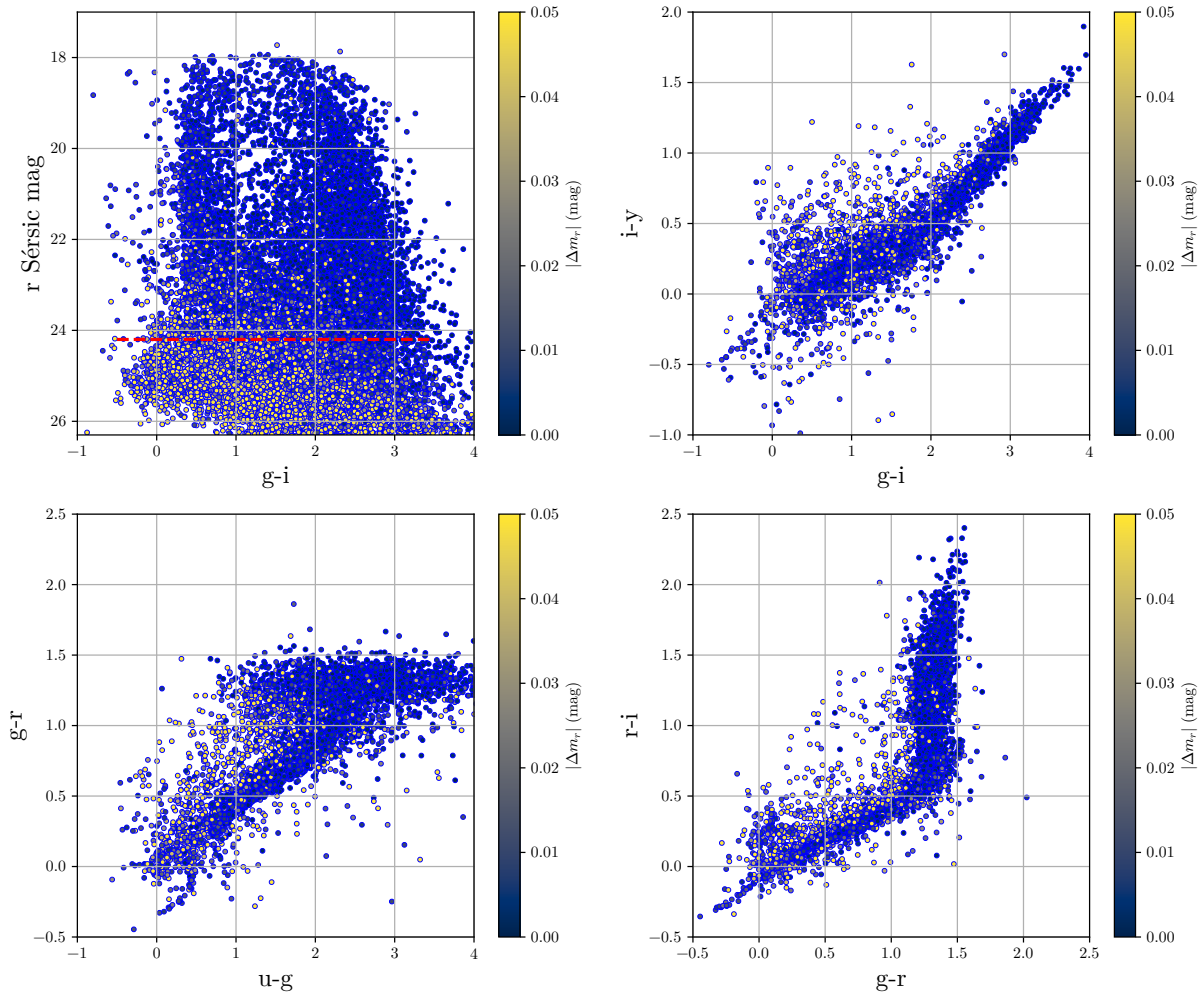


Figure 6. The distribution of objects classified as "stars" by HST in color-magnitude (top left) and color-color diagrams. The sample includes both resolved and unresolved classifications by Rubin. Only objects with $r < 24.2$ are shown in the three color-color diagrams. Symbols are color-coded by the absolute value of Rubin's Δm in the r band. The color-color distribution of objects with $|\Delta m| > 0.05$ mag (about 5% of all objects classified by HST as "stars" and brighter than $r = 23$) is inconsistent with the main stellar locus and similar to that expected for barely resolved or unresolved binary stars.

class in the r band. As evident, a significant fraction of HST sources within about two magnitudes from the Rubin faint limit are misclassified when using default `extendedness` parameter. Misclassifications have a more severe impact on candidate stellar sample: only 26% of "stars" selected by `extendedness`=0 condition are unresolved in HST data. This high level of stellar sample contamination by faint galaxies increases towards the faint end (see the bottom left panel in Figure 5). As the same time, the selection completeness for "stars" (objects unresolved by HST) is only 67%. Due to much higher counts at faint magnitudes and high Galactic latitudes, the purity and completeness for galaxies are much higher: 98% and 89%, respectively. In other words, a simple classifier that declares every object to be a galaxy would formally perform quite well. Therefore, we will focus our analysis in the next Section on the classifier's performance when selecting unresolved objects.

2.5. *Suspect HST Classification Labels*

There is a subset of objects classified as "stars" (unresolved) by HST which appear clearly resolved when using Rubin's default `extendedness` parameter. Figure 6 shows their distribution in color-color diagrams,

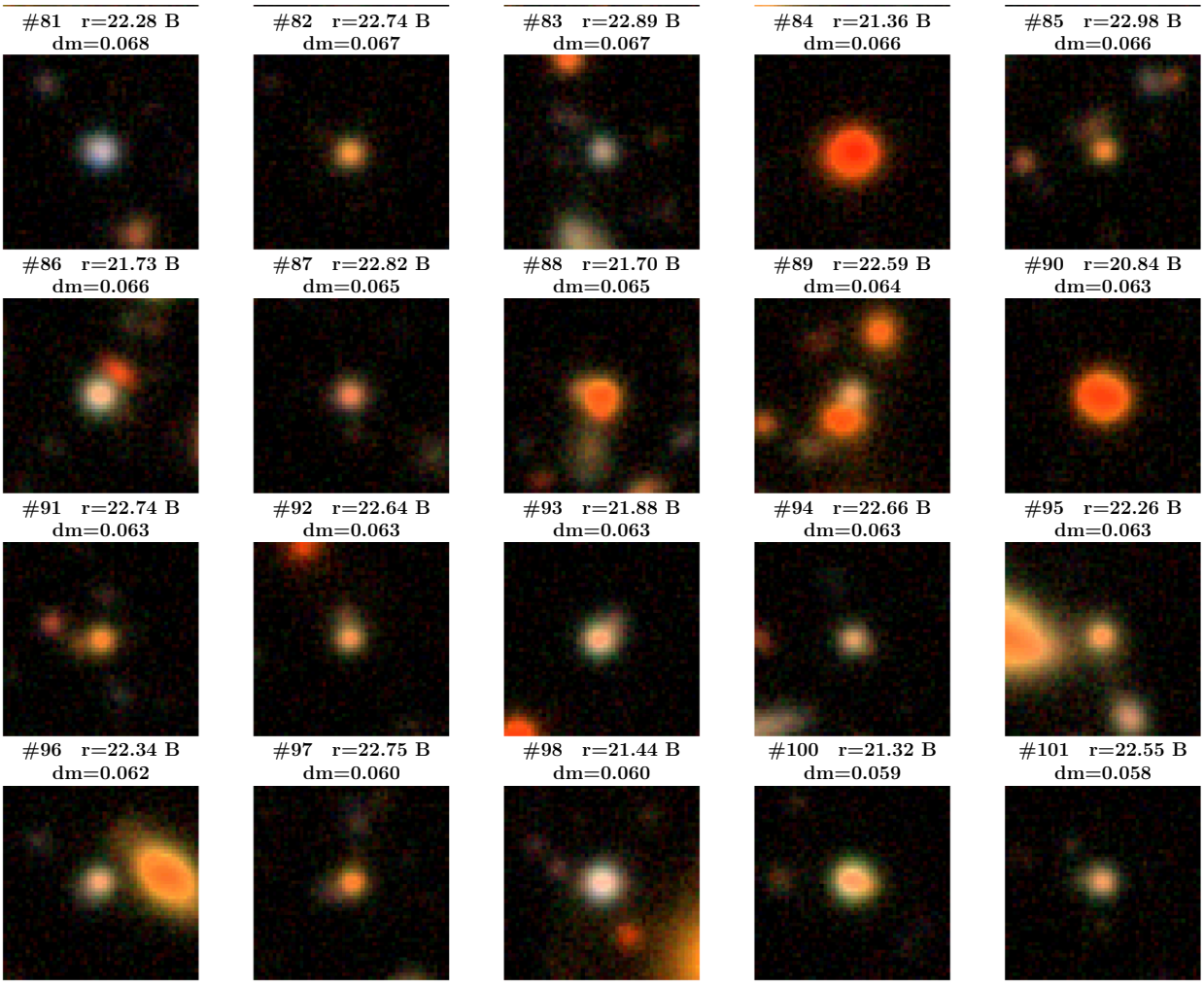


Figure 7. The Rubin *gri* color composite images extracted from "deep_coadd", 10 arcsec by 10 arcsec (50x50 pixels), for a random subsample of 20 (out of 443) HST-classified stars selected using $r_{\rm CModel} < 23$ and $m_{\rm PSF,r} - m_{\rm CModel,r} > 0.05$ mag. About 98.6% of 443 objects are detected as blended objects.

with symbols color-coded by Δm in the r band. As evident, their color-color distribution is inconsistent with main stellar locus and similar to that expected for barely resolved or unresolved binary stars (see Figures 2 and 4 in V. Smolčić et al. 2004.) To test this hypothesis, we visually inspected several hundred images (443 objects with HST label "star", $r < 23$ and $\Delta m > 0.05$ mag; see Figure 7 for an example). Only 1.4% of selected objects are detected as "isolated" (computed using processing flag "detect_isIsolated") and about half seem to have a nearby stellar-like companion. In some cases a bright source or an extended galaxy is nearby, and about one tenth or fewer appear bluish and consistent with quasars. Detailed examination of such cases is beyond the scope of this work.

Because we aim to test statistical performance of our star-galaxy separation method on unambiguous objects, we exclude objects with HST "star" label that both appear resolved in Rubin data and also have Rubin colors inconsistent with the main stellar locus.

We checked whether HST-labeled stars lie near the empirical main stellar locus in Rubin colors quantified using data for stars with $r < 23$ from the COSMOS field. For each source, we calculated its shortest distance to the locus using five dereddened CModel colors: $(u-g, g-r, r-i, i-z, z-y)$. Because the scatter in these colors changes with magnitude, we set a separate distance threshold in each Rubin r-band CModel magnitude bin. Each threshold was set to include at least 80% of HST-labeled stars with measurements in all five colors.

We excluded an HST-labeled star from the performance sample only when it was both beyond the color threshold and classified as resolved by Rubin in the r band (extendedness$_r = 1$). This removed 836 of the 15,760 HST-labeled stars (5.30%), leaving 14,924 (94.70%). In the two magnitude ranges of primary interest for measuring performance, the exclusions were 104 of 1,388 stars (7.49%) at $24 < r < 24.5$ and 116 of 1,700 stars (6.82%) at $24.5 < r < 25$. Stars without the measurements needed to check both criteria were not excluded by this procedure.

3. STAR-GALAXY SEPARATION METHODOLOGY

We first briefly review the Bayesian star-galaxy separation method proposed in SIL2020 and then apply it to single-band Rubin data, followed by its extension to the multi-band case. Our principal measured object property is the difference between PSF, $m_{\rm PSF}$, and CModel, $m_{\rm CModel}$, magnitudes, $\Delta m = m_{\rm PSF} - m_{\rm CModel}$. For unresolved sources, this Δm is consistent with zero within the measurement noise, while for resolved objects $\Delta m > 0$ (see the top left panel in Figure 3). A detailed comparison and statistical discussion of this quantity and several other quantities proposed for star-galaxy separation in the literature is available in SIL2020.

3.1. *Bayesian Star-Galaxy Separation Method from SIL2020*

We first focus on the single-band case and then extend our discussion to the multi-band case. Using realistic simulations, SIL2020 showed that adopting a fixed, predefined threshold for Δm to select resolved objects is not optimal. The threshold should depend on magnitude because of variations in the object populations and signal-to-noise ratio, as well as on sky location because the galaxy-to-star count ratio varies across the sky. Moreover, for an optimal combination of information from different bands or from other classification methods (e.g., based on colors), the observed Δm distribution needs to be transformed into a proper probability density function.

The subsample of objects with HST labels can be used to quantify the Δm distribution separately for unresolved and resolved objects (see Figure 8). However, such a "training" sample is currently available only for a very small fraction of the LSST footprint (less than 1% of the area). Instead, we use the fact that very large numbers of objects remaining even after binning in both sky location and magnitude can robustly constrain the two components of the Δm distribution even when such labels are not available a priori. We only use the subsample of objects with HST labels to independently assess the performance of our method.

3.2. *Mixture Model for Δm vs. $m_{\rm CModel}$ diagram*

We assume that a sample is selected from a suitably chosen sky region over which the stellar and galaxy populations can be assumed to be statistically uniform. Over angular scales of several degrees, stellar counts vary much more rapidly than galaxy counts (large galaxy clusters are an exception, but not important for the subsequent discussion). Based on the analysis of state-of-the-art Milky Way stellar population models by L. Palaversa et al. (2025), a sky region of several square degrees, corresponding to Rubin tracts[6] is sufficiently small for this purpose (and consistent with our choice of the COSMOS field for quantitative analysis).

Given a "tract" catalog, at a fixed CModel magnitude (see below for binning details), the distribution of Δm in a given band (with the band index suppressed for notational simplicity) is modeled as a mixture of an unresolved (star) component and a resolved (galaxy) component,

$$p(\Delta{\rm m}\,|\,m) = w_U\, f_{\rm unr}(\Delta{\rm m}) + w_R\, f_{\rm res}(\Delta{\rm m}), \tag{1}$$

where $w_U + w_R = 1$, and $f_{\rm unr}$ and $f_{\rm res}$ are properly normalized probability density functions. With this decomposition, the probability that an object is unresolved is given by Bayes' theorem as

$$p_S(\Delta{\rm m}) \equiv p({\rm unr}\mid \Delta{\rm m}) = \frac{w_U f_{\rm unr}(\Delta{\rm m})}{w_U f_{\rm unr}(\Delta{\rm m}) + w_R f_{\rm res}(\Delta{\rm m})}. \tag{2}$$

[6] In the Rubin Science Pipelines, tracts and patches are the two levels of spatial subdivision used for coadded images and object catalogs. They are designed to make processing and storage scalable while preserving continuity across the sky. A tract is a relatively large region of the sky, approximately 1.75° × 1.75° (the exact size depends on the sky map projection). that is processed as a unit. Tracts overlap slightly with neighboring tracts to avoid edge effects. Each tract is subdivided into a regular grid of patches, with each tract in DP2 containing 81 patches. The patch size is about 12 arcmin × 12 arcmin (approximately, one CCD).

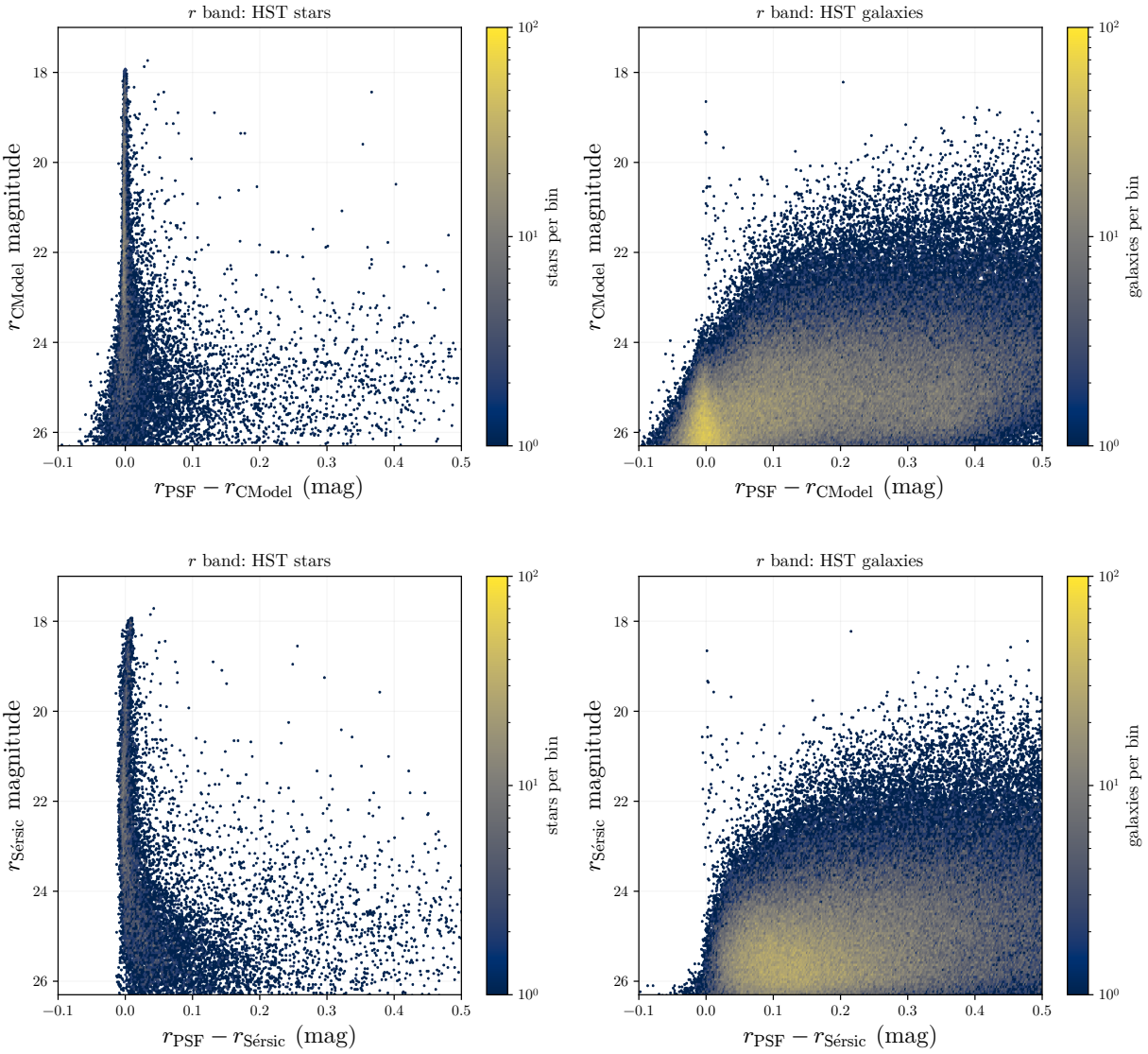


Figure 8. The panels compare the Δm vs. $m_{\rm CModel}$ (top row) and the Δm vs. $m_{\rm Sersic}$ (bottom row) distributions for unresolved (left) and resolved (right) objects classified using HST labels. Note that some objects in the left panels appear clearly resolved by Rubin (Δm > 0.05), even for $r < 24$ (see §2.5). Galaxies dominate the region around Δm ~ 0 at the faint end ($r > 24$) much more strongly in the top right panel than in the bottom right panel. Quantitative modeling of the two-dimensional distributions in the bottom row is at the core of the star-galaxy separation method discussed here.

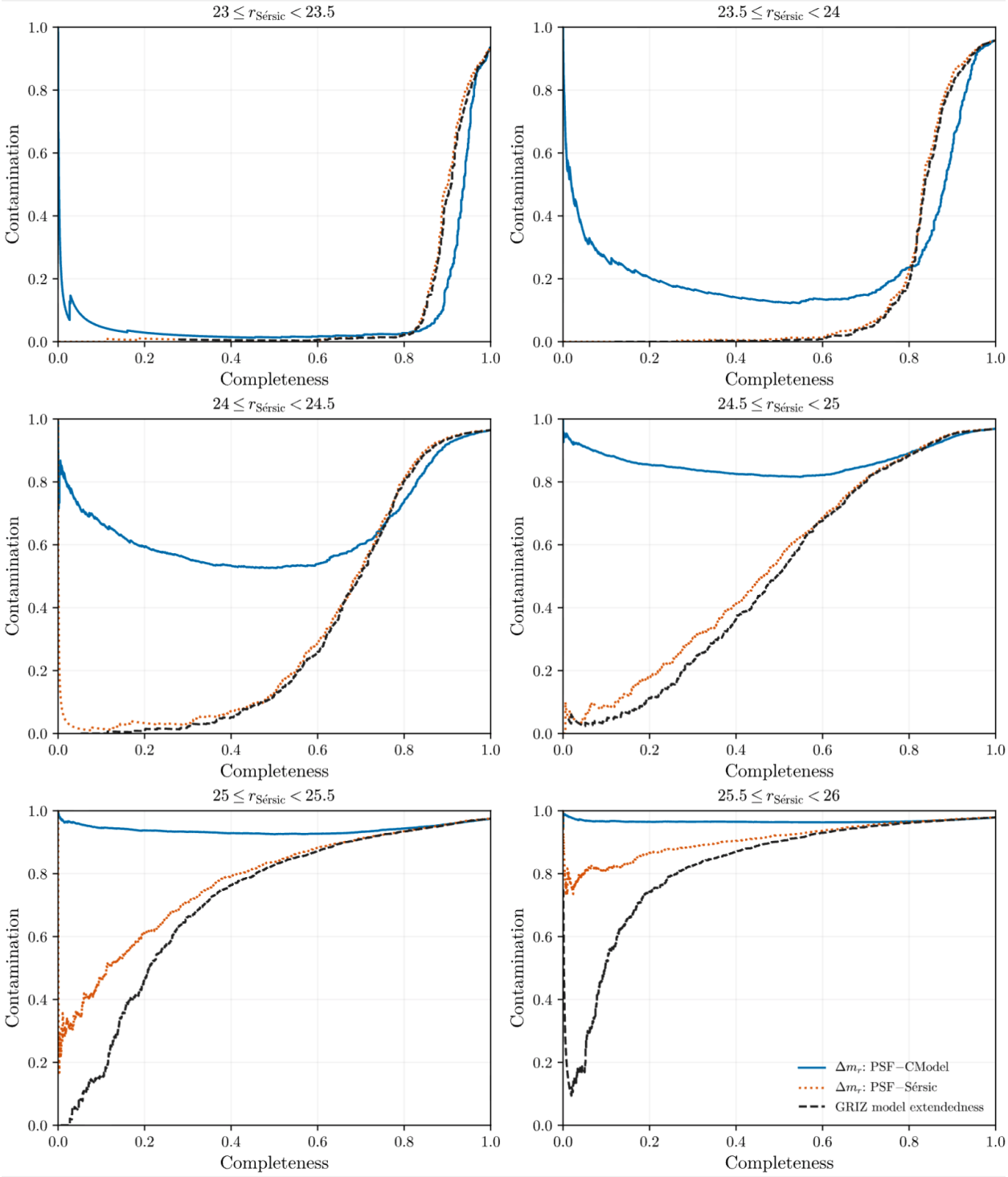


Figure 9. A comparison of contamination vs. completeness curves for unresolved sources using two versions of the r-band Δm based on Sersic and CModel fluxes, in different magnitude bins, as marked. Note much lower levels of contamination when Sersic fluxes are used, especially at the faint end.

and the probability that it is resolved is $p_G = 1 - p_S$ (note that we use indices S and unr, or G and res interchangeably).

The same model predicts the performance, completeness and contamination, for each class as a function of a hard Δm threshold. If $\Delta\mathrm{m} < \Delta\mathrm{m}_{\rm cut}$ is classified as unresolved (S),

$$\mathrm{Completeness}(S) = F_{\rm unr}(\Delta\mathrm{m}_{\rm cut}) = \int_{-\infty}^{\Delta\mathrm{m}_{\rm cut}} f_{\rm unr}(\Delta\mathrm{m}) d\Delta\mathrm{m}, \tag{3}$$

and

$$\mathrm{Contamination}(S) = \frac{w_R F_{\rm res}(\Delta\mathrm{m}_{\rm cut})}{w_U F_{\rm unr}(\Delta\mathrm{m}_{\rm cut}) + w_R F_{\rm res}(\Delta\mathrm{m}_{\rm cut})}. \tag{4}$$

We test these predictions below using HST-based labels.

Therefore, a good model for $p(\Delta\mathrm{m}\,|\,m)$ (eq. 1) is crucial. Informed by observed Δm distributions (see Figure 8), we model the unresolved component, $f_{\rm unr}$, as a single Gaussian with mean μ_U, width σ_U and weight w_U. The resolved component is modeled as a sum of K skew-normal distributions with weight $w_{R,k}$, location $\mu_{R,k}$, scale $\sigma_{R,k}$, and shape parameter $\alpha_{R,k}$. The full mixture probability density function is

$$p(\Delta m\,|\,m) = w_U\,\mathcal{N}(\Delta m|\,\mu_U,\,\sigma_U) + \sum_{k=1}^{K} w_{R,k}\,\mathcal{SN}(\Delta m|\,\mu_{R,k},\,\sigma_{R,k},\,\alpha_{R,k}), \tag{5}$$

where $\mathcal{N}$ denotes the Gaussian and $\mathcal{SN}$ the skew-normal distribution, and the weights are constrained to sum to unity: $w_U = 1 - \sum_k w_{R,k}$ (functions $\mathcal{N}$ and $\mathcal{SN}$ are properly normalized probability density functions). Here $\mathcal{SN}(x \mid \mu, \sigma, \alpha) = 2\phi(t)\Phi(\alpha t)/\sigma$, with $t = (x-\mu)/\sigma$, where ϕ and Φ are the standard-normal density and cumulative distribution functions. The galaxy-to-star count ratio, which serves as Bayesian prior, is

$$N_{GS} \equiv \frac{N_G}{N_S} = \frac{\sum_{k=1}^{K} w_{R,k}}{w_U}. \tag{6}$$

The expression in eq. 2 can be recast as

$$p_S(\Delta\mathrm{m}) = \frac{1}{1 + N_{GS}\,BF}, \tag{7}$$

where the Bayes factor (the likelihood ratio for the galaxy and star hypotheses) is

$$BF = \frac{f_{\rm res}}{f_{\rm unr}}. \tag{8}$$

With $N_{GS} \sim 30$ (see Figure 14), it is sufficient to have $BF \sim 0.033$ for p_S to drop below 0.5 (because the sample is highly imbalanced). This decomposition in terms of prior and Bayes factor is important later when combining multiple sources of classification information (with the overall Bayes factor equal to the product of individual Bayes factors).

3.2.1. *Fitting Details*

The fitting is performed independently in bins of CModel magnitude. At bright magnitudes ($m_{\rm CModel} < 22$), where the resolved population is relatively simple, we use $K = 2$ resolved components in bins of width 1–3 mag spanning $15 < m_{\rm CModel} < 22$. At faint magnitudes ($m_{\rm CModel} \geq 22$), where the galaxy morphology distribution is more complex, we use $K = 3$ resolved components in finer bins of width 0.5–1 mag spanning $22 \leq m_{\rm CModel} < 26$. Within each magnitude bin, Δm values in the range $(-0.5, 1.5)$ are binned into 300 bins and normalized to unity. The mixture model is fit to these histograms using bounded least-squares optimization with Poisson-like uncertainties $\sigma_i = \sqrt{N_i + 1}\,/\,(N\,\delta)$, where N_i is the count in bin i, N is the total number of sources in the magnitude slice, and δ is the histogram bin width.

This fitting procedure is carried out in three steps. In the first step, all mixture parameters are fit independently in each magnitude bin. This free fit works well at bright magnitudes, where unresolved and resolved sources are clearly separated in Δm. However, at faint magnitudes ($m_{\rm CModel} \gtrsim 24$), photometric

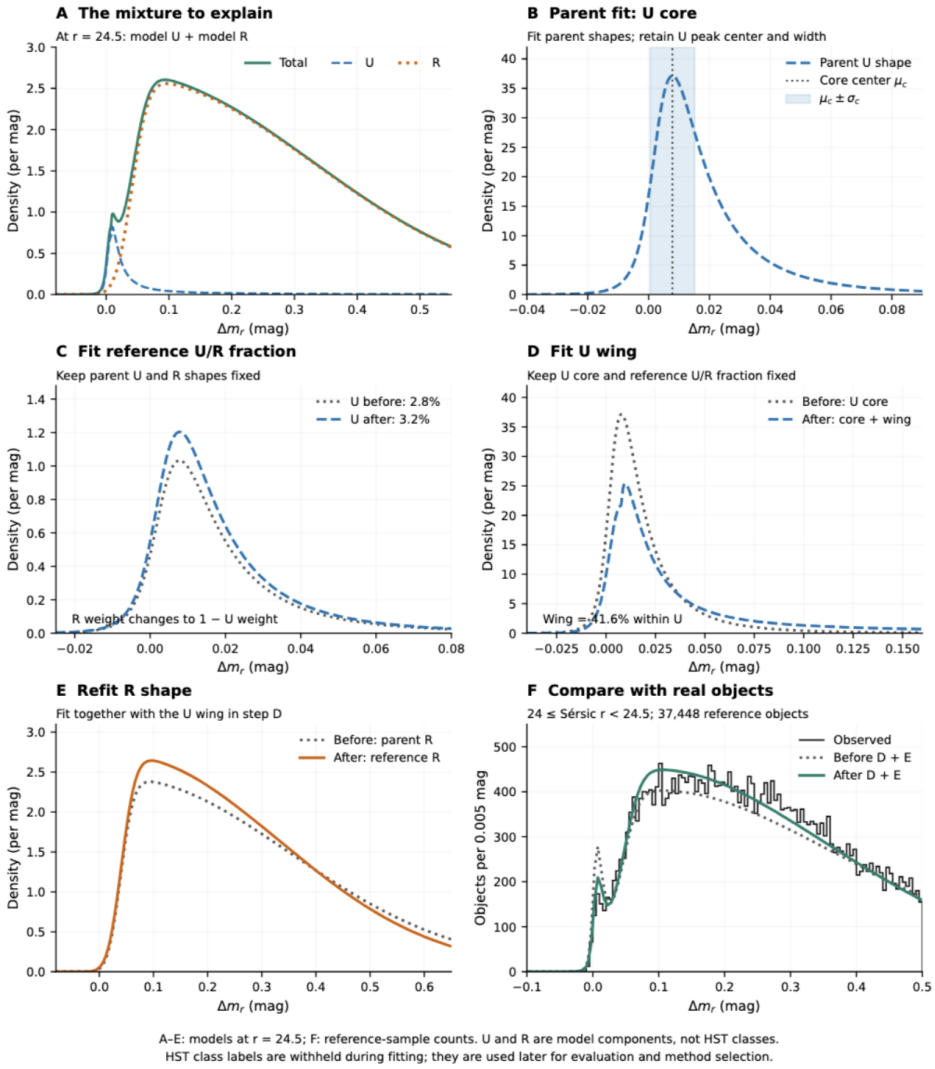


Figure 10. V12 fitting details. DANIEL ADD CAPTION!

noise broadens the unresolved Gaussian until it becomes degenerate with the resolved components. The fitter can no longer reliably distinguish the two populations, causing the unresolved Gaussian to absorb flux from the galaxy distribution. This inflates the inferred star fraction at faint magnitudes and the turn-over of the galaxy-to-star count ratio at the faint end, see Figure 14.

To prevent this degeneracy, we impose smooth constraints on the unresolved component. In the second step, the per-bin star parameters from step 1 are replaced with smooth functions of magnitude: μ_U is fixed to a single value (the median across all bins, typically ≈ 0 in each band); $\sigma_U(m)$ is fit with the photometric error model from B. Sesar et al. (2007) (their eq. 5),

$$\sigma_U(m) = \sqrt{\sigma_0^2 + a^2 \cdot 10^{\,b\,(m-25)}}, \tag{9}$$

where σ_0, a, and b are free parameters; and $\log_{10} w_U(m)$ is modeled as a linear function of magnitude, using only bins brighter than $m_{\mathrm{CModel}} = 24$ where the free fit is reliable. This smooth model is then extrapolated to fainter magnitude bins. In the third step, with the unresolved component (μ_U, σ_U, w_U) fixed to the smooth curves from step 2, only the resolved skew-normal parameters are re-fit in each magnitude bin. The resolved weights are rescaled to sum to $(1 - w_U)$ so that the total mixture remains a proper probability density. The effectiveness of this constrained fitting procedure is demonstrated in Figure **??**, which compares the

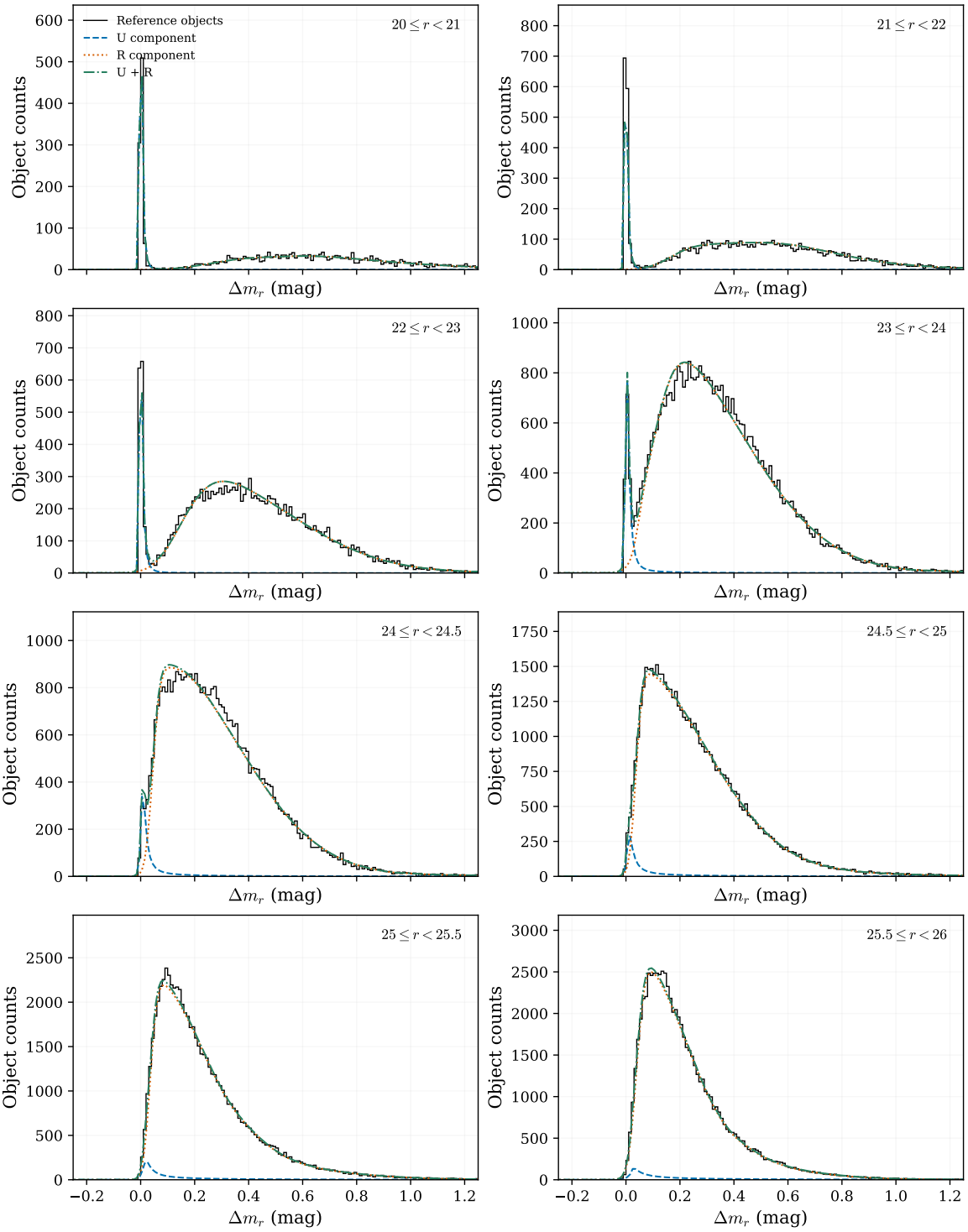


Figure 11. Observed r-band Sérsic-based Δm histograms (black) and the best-fit two-component mixture models (dashed: unresolved component; dotted: resolved component) in magnitude bins, as marked.

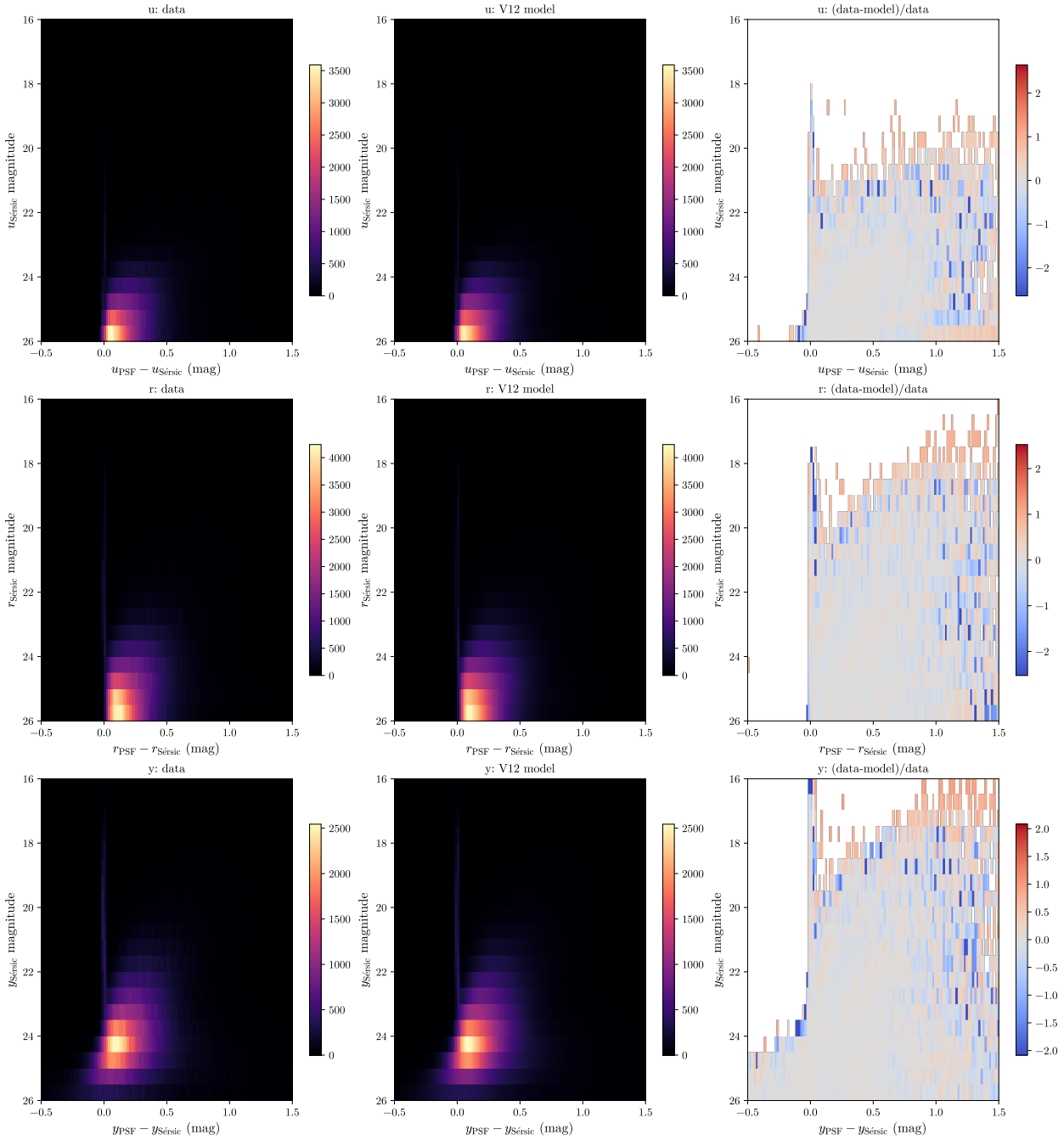


Figure 12. Two-dimensional comparison of the observed (left), modeled (center), and fractional residual (right) source counts in the Δm vs. $m_{\rm Sersic}$ plane, for the u, r, and y bands from top to bottom. Residuals show $(N_{\rm data} - N_{\rm model})/N_{\rm data}$. Results are similar other bands.

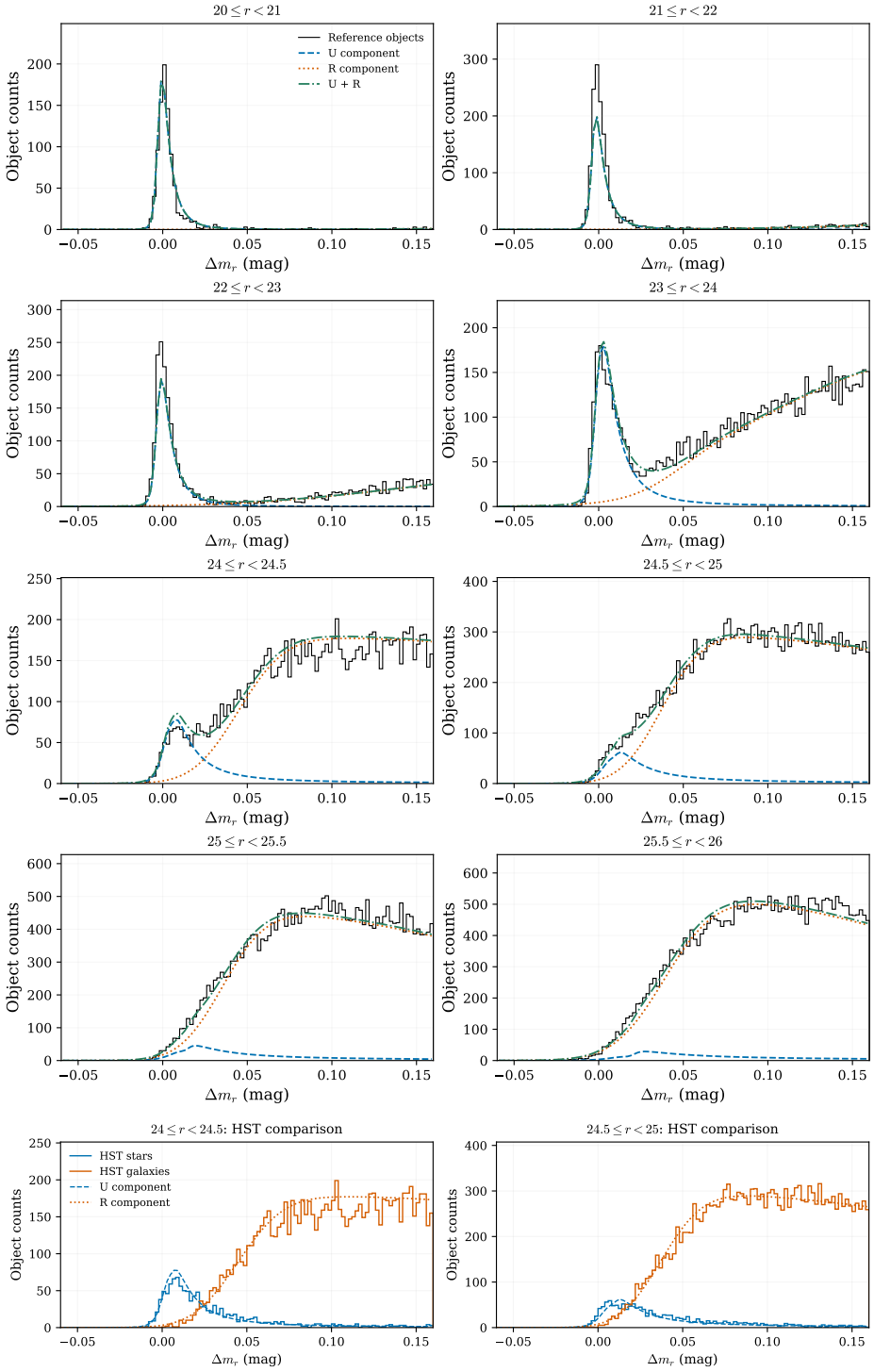


Figure 13. The top four rows show zoomed-in comparison of observed r-band Δm distributions and their best-fit two-component models, in magnitude bins, as marked. Black histograms show the observed counts, while dashed blue, dotted orange, and dot-dashed green curves show the unresolved and resolved components and their sum, respectively. The bottom row compares the two best-fit model components (dashed and dotted smooth lines) to observed distributions for two subsets of source classified using HST labels, in the $24 \leq r \leq 24.5$ and $24.5 \leq r \leq 25$ magnitude bins. Note the excellent agreement between observed and modeled distributions.

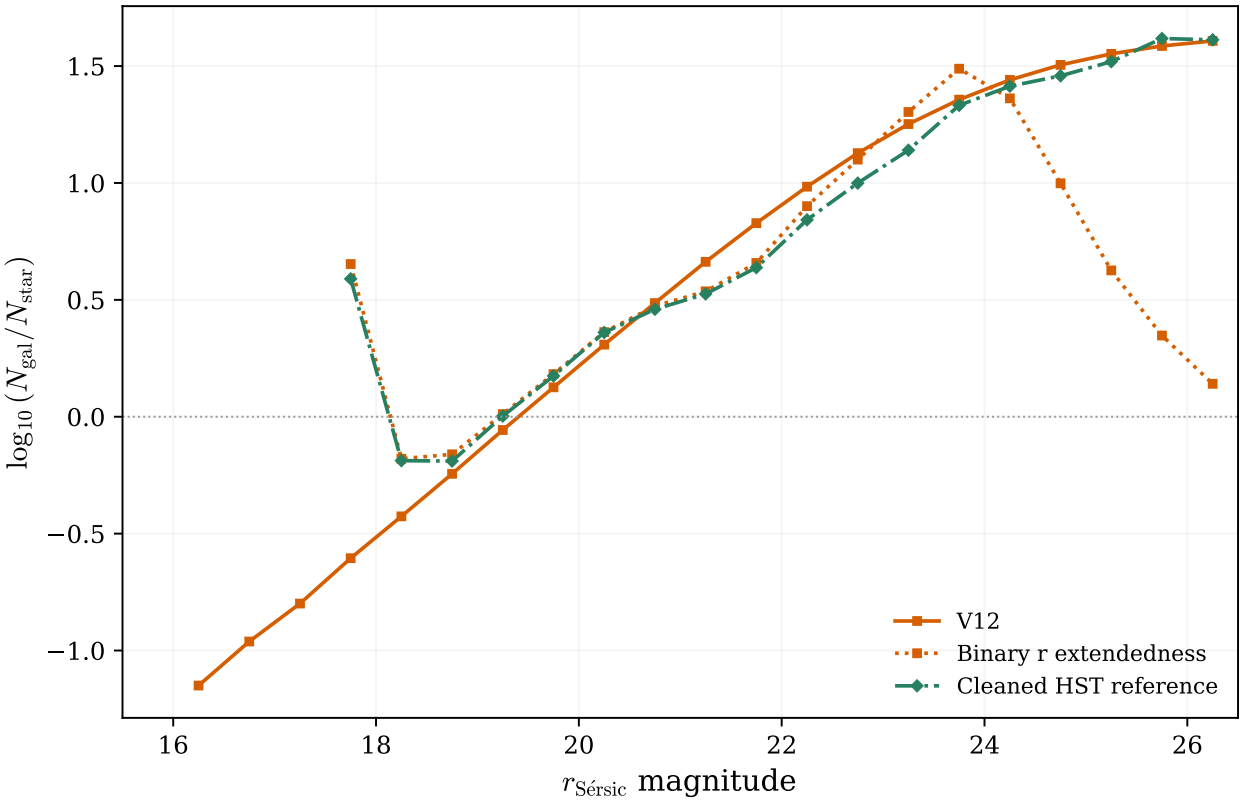


Figure 14. A comparison of model-predicted (solid red line) and HST-observed (green dot-dashed line) galaxy–to-star count ratio as a function of the r-band Sersic magnitude. The counts of sources classified using the r band `extendedness` (dotted line) shows a spurious turnover at $r > 24$ due to a large number of galaxies misclassified as stars.

galaxy-to-star count ratio $N_{\rm gal}/N_{\rm star}$ as a function of magnitude for models with and without the smooth constraints. The unconstrained model exhibits a spurious turnover at faint magnitudes, while the constrained model recovers the expected monotonically increasing ratio in agreement with the COSMOS2020 external truth labels (note that in this sample at the faint end galaxies outnumber stars by more than a factor of 30).

The model parameters are selected according to the CModel magnitude bin in which each source falls. The quality of the fits is illustrated in Figure 11, which shows the mixture model overlaid on the observed Δm histograms in each magnitude bin, and in Figure 12, which shows the 2D data, model, and residual maps in the Δm vs. $m_{\rm CModel}$ plane.

3.3. *The Performance of the Δm Mixture Model*

Here we use HST labels to evaluate completeness and contamination for the stellar sample and compare them with the mixture-model predictions. We focus on the stellar subsample and the "transition" magnitude range, $24 < r < 25$, in two intervals: $24 \leq r < 24.5$ and $24.5 \leq r < 25$. Brighter than this range, population mixing is much smaller and measurements more accurate, leading to more accurate star-galaxy separation, whereas at fainter magnitudes the performance deteriorates rapidly (especially for the stellar subsample).

Using the mixture model from eq. 1, we compare completeness and contamination for the stellar subsample predicted by eqs. 3 and 4 with direct counts from the cleaned HST sample for $24 \leq r < 24.5$ and $24.5 \leq r < 25$ in Figure 15. As the two top panels show, the model underpredicts observed contamination levels at every completeness level, and in both magnitude bins. For example, at the 50% completeness level, the model predicts negligible contamination while the observed contamination exceeds 40%.

This large discrepancy between predicted and observed contamination can be traced to the behavior of best-fit resolved mixture component at $\Delta m \sim 0$ As the two top panels in Figure 13 show, the best fit

for resolved component (dotted line) does not extend approximately linearly from larger values of Δm as suggested by... ???

WE NEED TO ADD ANOTHER ROW TO Figure 13, with two histograms for HST stars and galaxies XXXXXDONE!.

A smooth extrapolation of the resolved component from $\Delta m > 0.04$ to $\Delta m = 0$ (which corresponds to 50% completeness level for unresolved subsample) implies contamination levels consistent with the observed ones ($\sim$40% and $\sim$80%).

XXX ZI stopped here. We need more work on models to finish this subsection... WHY DO NEW MODELS FIX RESOLVED COMPONENT?

WE NEED TO DISCUSS THIS MODEL AGAIN – THE ACTUAL PROBLEM COMES FROM RESOLVED COMPONENT BUT THIS MODEL CHANGES UNRESOLVED COMPONENT

To investigate this discrepancy, we also test a core-plus-tail model for the unresolved component.

$$U_{\rm cpt}(\Delta m_r \mid m) = A\left[(1-\epsilon)G(\Delta m_r \mid m) + \epsilon\,(G * E_\tau)(\Delta m_r \mid m)\right]. \tag{10}$$

Here G is a Gaussian with center and magnitude-dependent width fixed by an earlier r-only calibration; E_τ is an exponential distribution of nonnegative offsets with mean τ; and $*$ denotes convolution. The total unresolved fraction is A, of which ϵ belongs to the tail. We also refit the resolved density, which integrates to $1 - A$ and can be nonzero near $\Delta m_r = 0$ (Figure 13). In each interval, we minimize the mean negative log-likelihood of the r photometry plus $0.001\,A\epsilon$.

In both target intervals, the core-plus-tail model gives closer contamination predictions at $p_S > 0.2$: it predicts 49.4% and 64.8%, and HST measures 53.9% and 76.4% for its selections. The absolute prediction differences fall from 37.0 to 4.5 percentage points in the brighter interval and from 30.2 to 11.6 percentage points in the fainter interval. Its predicted completeness is also closer to HST at this threshold.

3.4. *The Multi-band Case*

We expect classification to improve when Δm measurements from multiple bands are used. When combining information from different bands, a different approach is required than when combining multiple measurements of the same type from a single band. The main reason is that it is sufficient for a source to be reliably identified as resolved in just one band, irrespective of in how many other bands it was unresolved. Therefore, combining information from different bands is a nontrivial statistical problem that is made more complex by astrophysical considerations. For example, galaxy morphology is wavelength dependent: different bands emphasize different stellar populations, dust distribution, star-forming regions, bulges, and disks, and thus a galaxy can show strong resolved structure in one band while being close to a PSF in another. A decisive measurement in one band should therefore not necessarily be diluted by several bands with weak or absent evidence. Consequently, combining measurements from multiple bands is not as straightforward as combining multiple measurements of Bayes Factors (see eq. 8) from individual bands using eq. 7 (see Section 5 in SIL2020 for details).

An optimal Bayesian approach would be to generalize eq. 1 to a vector form and model the true object class as an underlying latent variable. The multi-band measurements are correlated projections of this latent structure rather than independent measurements of one scalar size. We explored a Gaussian copula construction for combining single-band probabilities, but it did not provide a satisfactory description of the desired behavior. We therefore adopt a different approach: first combine the measured Δm values across bands into a single statistic and then derive classification criteria from its distribution, analogously to the single-band case.

Motivated by the fact that the multi-band Δm distribution for the unresolved component can be described as a multivariate Gaussian with diagonal covariance matrix, we define a standardized variable in each band

$$D_i = \frac{\Delta{\rm m}_i - \mu_{U,i}}{\sigma_{U,i}}, \tag{11}$$

where i is the band index and μ_U and σ_U are the best-fit parameter values for the Gaussian fit to the unresolved component (recall that $\mu_U \sim 0$). The utilized bands may be all of *ugrizy* or any subset of N

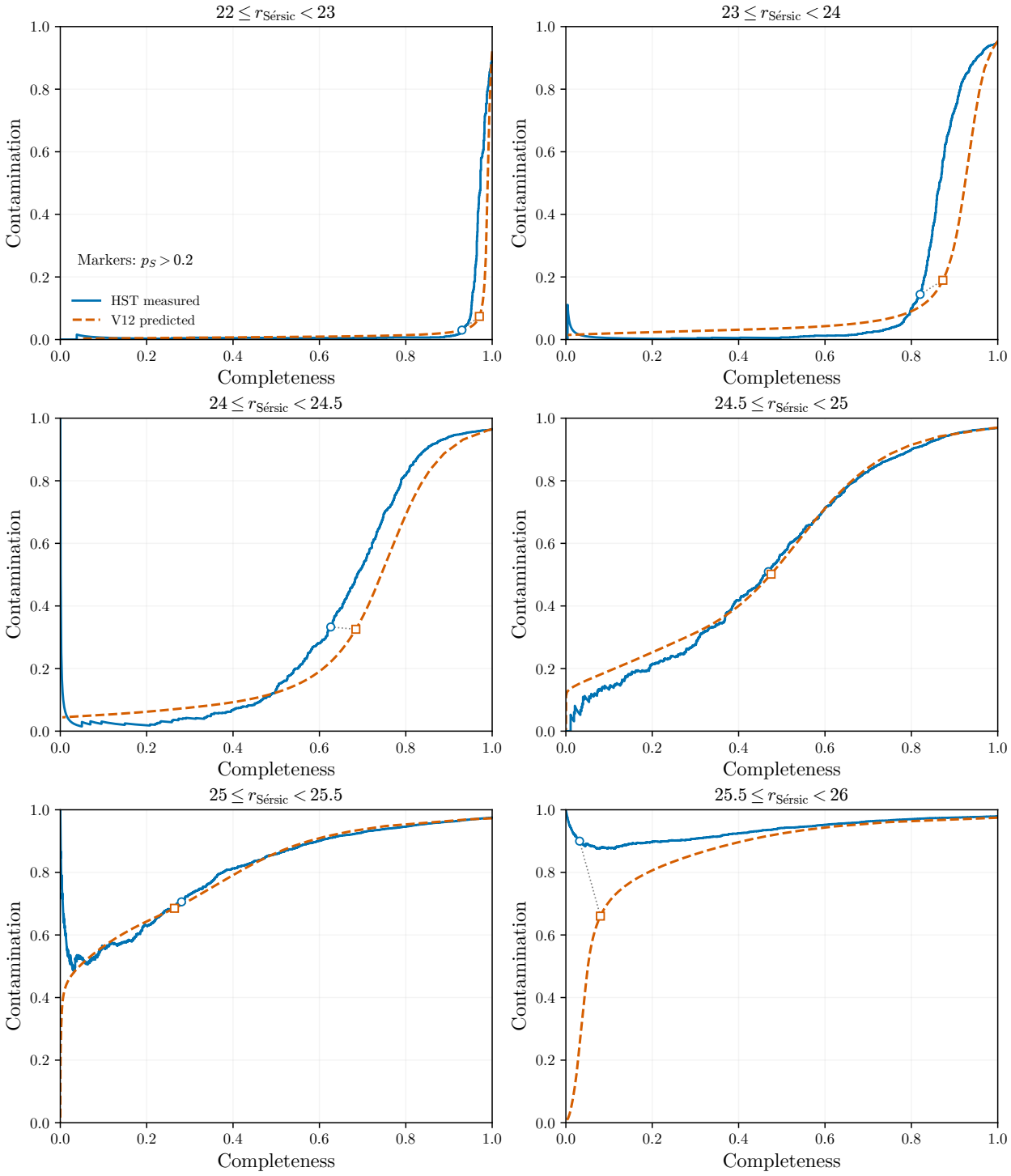


Figure 15. Model-predicted (red dashed lines) and HST-observed (solid blue lines) r-band completeness–contamination relations in different magnitude bins, as marked. Observed curves are based on direct HST counts of unresolved and resolved sources. The model-predicted curves are based on integrals given by eqs. 3 and 4. Open circles and squares mark the HST and model results for $p_S > 0.2$, respectively; dotted segments connect results at the same threshold. Selected samples are highly imbalanced and dominated by galaxies.

usable (depending on data depth and spectral energy distribution of objects of interest) bands, B. We then

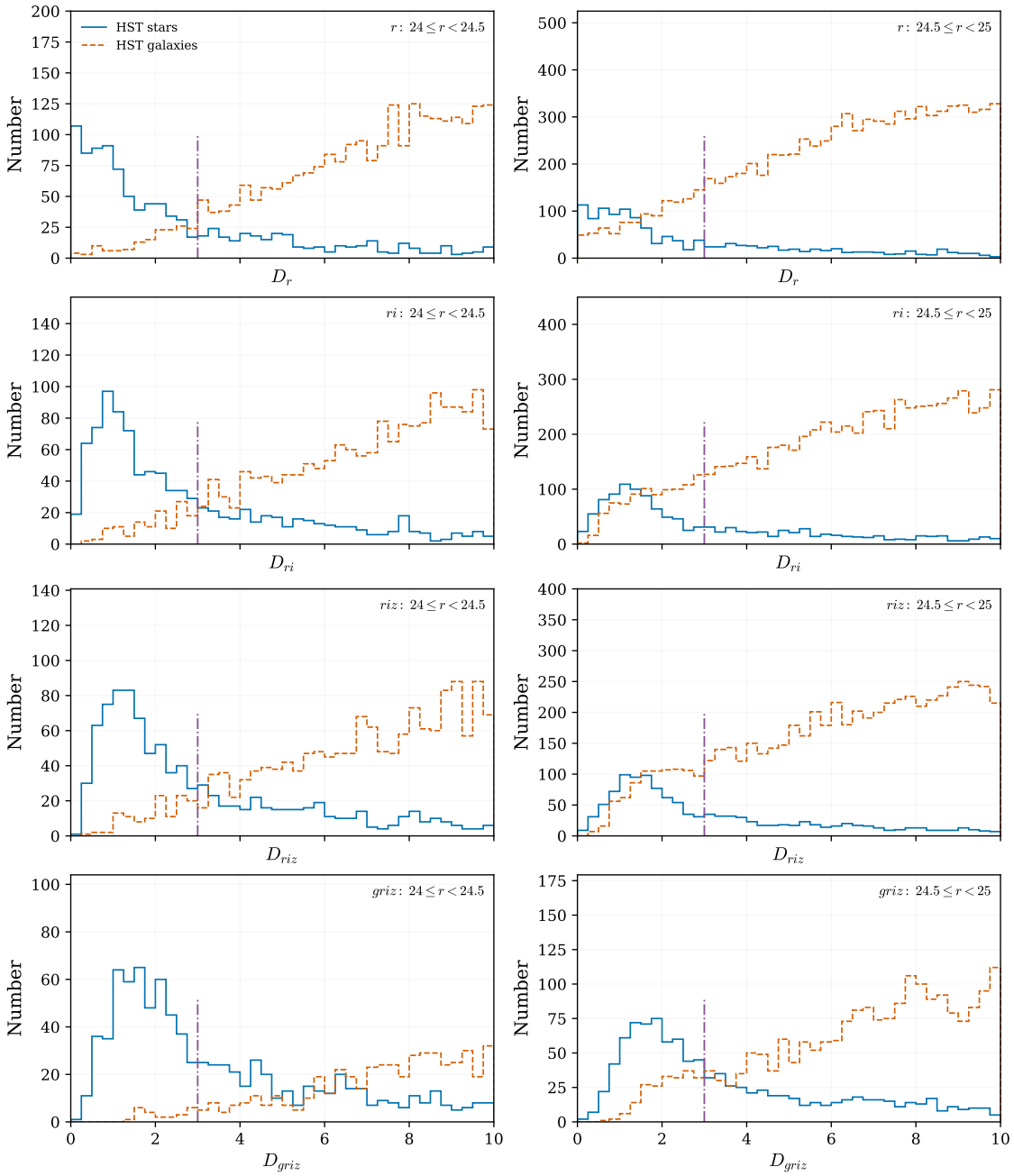


Figure 16. Observed D distributions of HST stars (blue) and HST galaxies (red). Rows show r, ri, riz, and $griz$ band combinations of the D metric; columns show $24 \leq r < 24.5$ (left) and $24.5 \leq r < 25$ (right). Histograms show object counts in bins of width $\Delta D = 0.25$, over $0 \leq D \leq 10$ (distributions for HST galaxies extend to much larger D values of ∼100). Half-height vertical lines mark the theoretical χ_N 95% thresholds: $D = 1.960$, 2.448, 2.795, and 3.080 for the four rows, respectively.

define a radial metric

$$D = \left(\sum_B D_i^2 \right)^{1/2} . \tag{12}$$

When $N = 1$, $D \sim |\Delta \mathrm{m}/\sigma_U|$, and it measures the distance from the origin in units of σ_U. For the unresolved component, D follows a χ-distribution with one degree of freedom, χ_1 (which is also the half-normal distribution). If μ_U cannot be ignored, D follows a noncentral χ distribution. When $N > 1$, for the unresolved

component $D \sim \chi_N$, with an expectation value approximately equal to $\langle D \rangle = \sqrt{N - 1/2}$. Depending on the desired completeness for the unresolved component, a hard D threshold of several times $\sqrt{N}$ can be used for sample classification (this is a standard case of hypothesis testing in frequentist statistics). However, without a quantitative description of the resolved component as well, the resulting sample contamination cannot be estimated.

Figure 16 shows D distributions for HST stars and HST galaxies, and their dependence on the r-band magnitude and band combination. The behavior of D distributions for HST stars is approximately consistent with expectation ($D \sim \chi_N$); for example, the fraction of HST stars selected by theoretical 95% D threshold is in the range XXX-XXX (and XXX-XXX for HST galaxies). The well-defined peak at $D < 3$ for HST galaxies implies that some HST galaxies appear truly unresolved in Rubin images. Only a lower limit on their fraction/number can be estimated by extrapolation of the observed distribution at $D > 3$ to smaller D values. For this reason, we do not pursue detailed fits to the observed D distributions and instead focus on estimating the completeness and contamination of selected candidate unresolved objects using HST truth labels, described next.

3.5. *Two-component Mixture Fit to D_{griz} Distributions*

Describe V12 fits, and completeness/contamination predictions

3.6. *The Performance of the D Metric-based Classification*

Here we again use HST labels to evaluate completeness and contamination for the stellar multi-band D-selected samples and compare them to the single-band performance. As before, we focus on the "transition" magnitude range, $24 \leq r < 25$, in two intervals, $24 \leq r < 24.5$ and $24.5 \leq r < 25$.

Figure 19 compares the empirical performance of the ri, riz and $griz$ band combinations to the r-band case, for a grid of D thresholds (for example, in the $griz$ case and $24 \leq r < 24.5$, a sample completeness in the range 20-50% can be obtained using D thresholds in the range 3.5 to 1.4). As more bands are added, the sample contamination decreases at D thresholds that correspond to completeness levels below 50%. For example, at a completeness level of 20%, contamination decreases from 40% to 30% between the r and $griz$ cases. In the fainter bin, improvement is smaller though still clearly present. In practice, detailed optimization of the tradeoff between sample completeness and contamination depends on science case.

3.7. *Using D Metric-based Classification with Target Completeness or Contamination*

3.8. *Comparison with `griz_model_extendedness`*

Comment that `griz_model_extendedness` adds fluxes directly, which is clearly wrong

3.9. *Improved Stellar Classification Using Color Selection*

As shown in preceeding sections, the performance of morphological separation of unresolved and resolved sources starts to rapidly deteriorate about 2 magnitudes brighter than the 5σ limiting magnitude for point sources. Selection of unresolved objects is especially problematic because the sample is highly imbalanced at the faint end. At the same time, samples of faint stars are valuable for studying the Milky Way structure (e.g. stellar streams in the halo). Here we investigate if color selection can be used to decrease contamination without significant decrease in completeness of such samples.

Figure 24 compares color distributions of morphologically selected samples analyzed in preceeding sections. As evident, HST galaxies (misclassified as stars) appear systematically bluer than stars, except for the $z - y$ color. Therefore, selection based on bluer colors can result in improved classification.

Motivated by color distributions shown in Figure 24, we investigate simple $g - r$ and $r - i$ selection criteria in Figure 25. The $g - r > 0.3$ cut was chosen as appropriate for selecting blue stars near the turn-off point on the principal stellar locus (composed of main sequence and red giant stars; see e.g., L. Palaversa et al. 2026). The cut $g - r > 1.0$ selects red (M type) stars and excludes almost all galaxies. The cut $r - i > 0.8$ is similar but more effective at excluding galaxies.

The top panel in Figure 25 shows substantial decrease of unresolved sample contamination when $g - r > 0.3$ condition is added, both in single-band and multiband cases. For example, for $24 \leq r < 24.5$ bin at fiducial

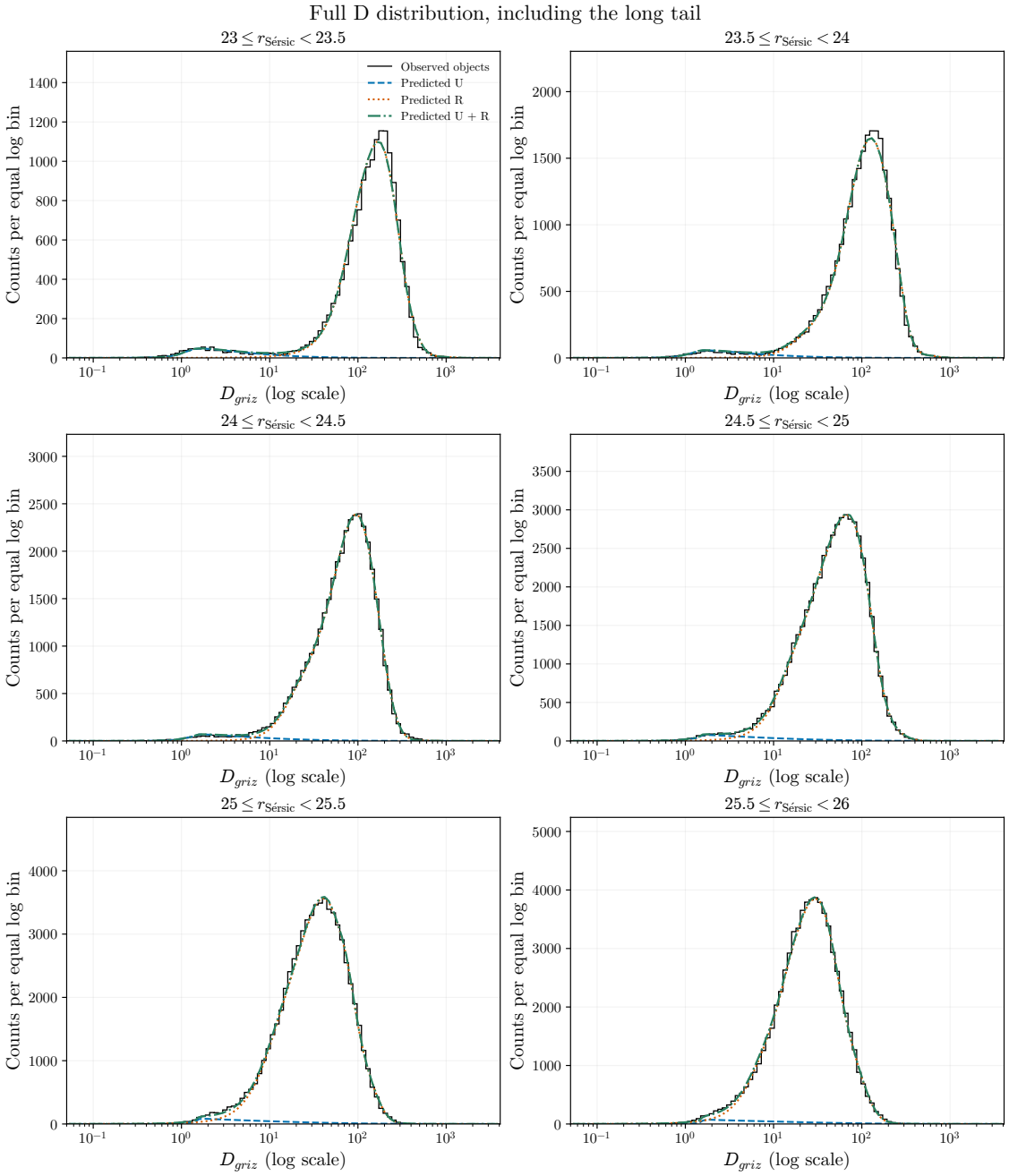


Figure 17. The best-fit two-component models (smooth lines) to observed (histograms) D_{griz} distributions in different magnitude bins, as marked. Note that x axis is shown on logarithmic scale. The large peak corresponds to resolved component; unresolved component peaks at $D_{griz} \sim 2$.

stellar completeness of 50%, D_{griz} and $g - r > 0.3$ selection decrease contamination to 25% from 45% in the r band case without a color cut; at stellar completeness of 20%, the contamination level is even more dramatically decreased from 40% to 12%. Corresponding values for $24.5 \leq r < 25$ are 78% to 63%, and 72% to 54%.

Decrease of contamination is much more substantial in case of red stars. For example, even in the fainter bin, $24.5 \leq r < 25$, the addition of $g - r > 1.0$ criterion results in contamination as low as 20% with substantial completeness of 45%. When completeness of 20% is acceptable (relative to all stars, not just the red subsample), contamination levels as low as 10% can be achieved.

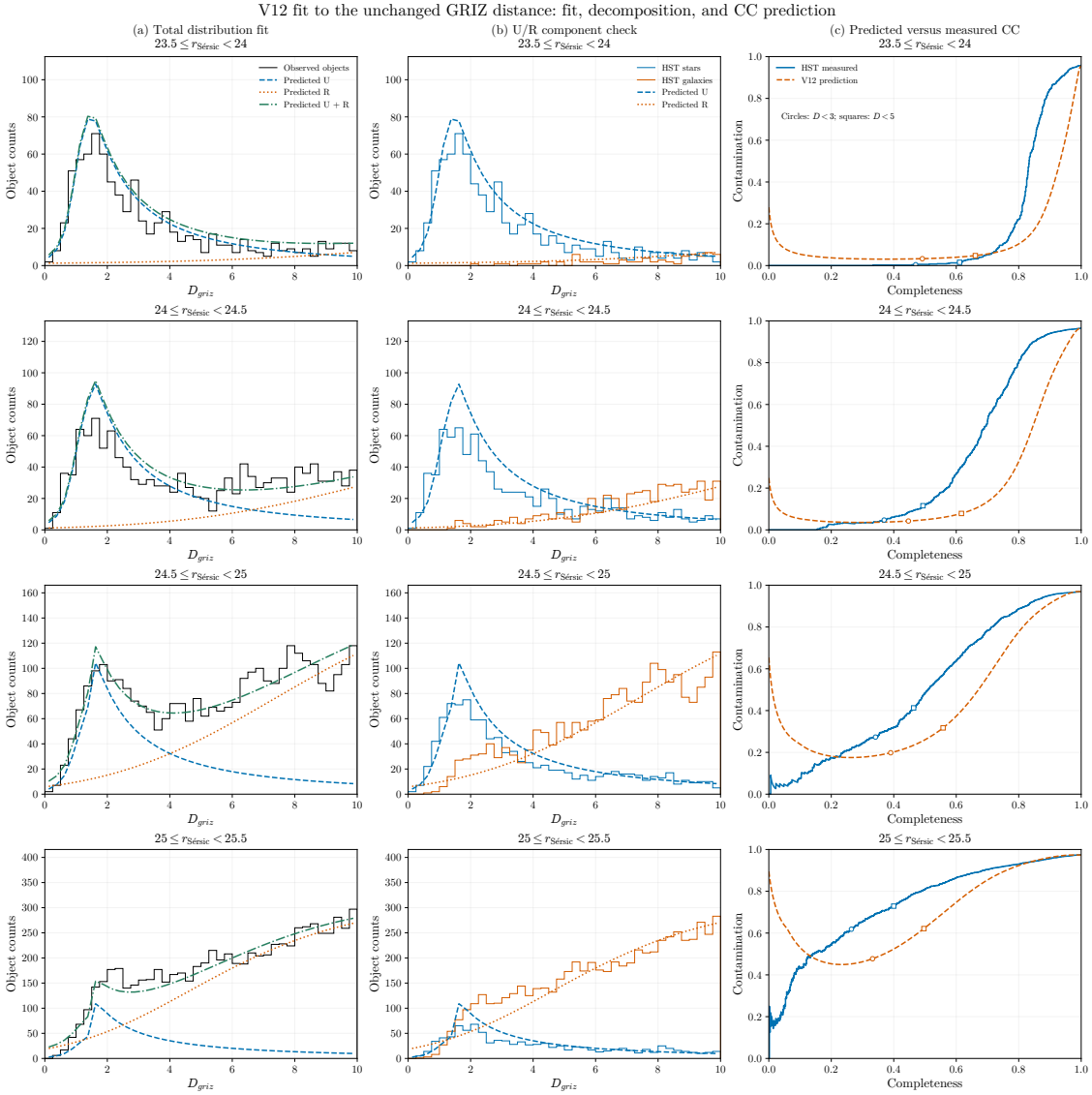


Figure 18. The panels in the first column are zoomed-in versions of Fig. 17, in four selected magnitude bins, as marked. Unresolved and resolved model components are shown with blue dashed and red dotted lines, respectively. The middle column is analogous to the left column, except that sources are separated by their HST labels (which were not used in model fitting). The right panels compare model-predicted and HST-observed contamination–completeness curves. The two symbols mark the locations of $D_{griz} = 3$ (circles) and $D_{griz} = 5$ thresholds.

3.10. *Comparison of Multiband D Metric to griz_model_extendedness Parameter*

`griz_model_extendedness` is well correlated with Δm. We need to compare how D_{griz} performance compares to `griz_model_extendedness` performance... But first ZI need to download data (`griz_model_extendedness`)

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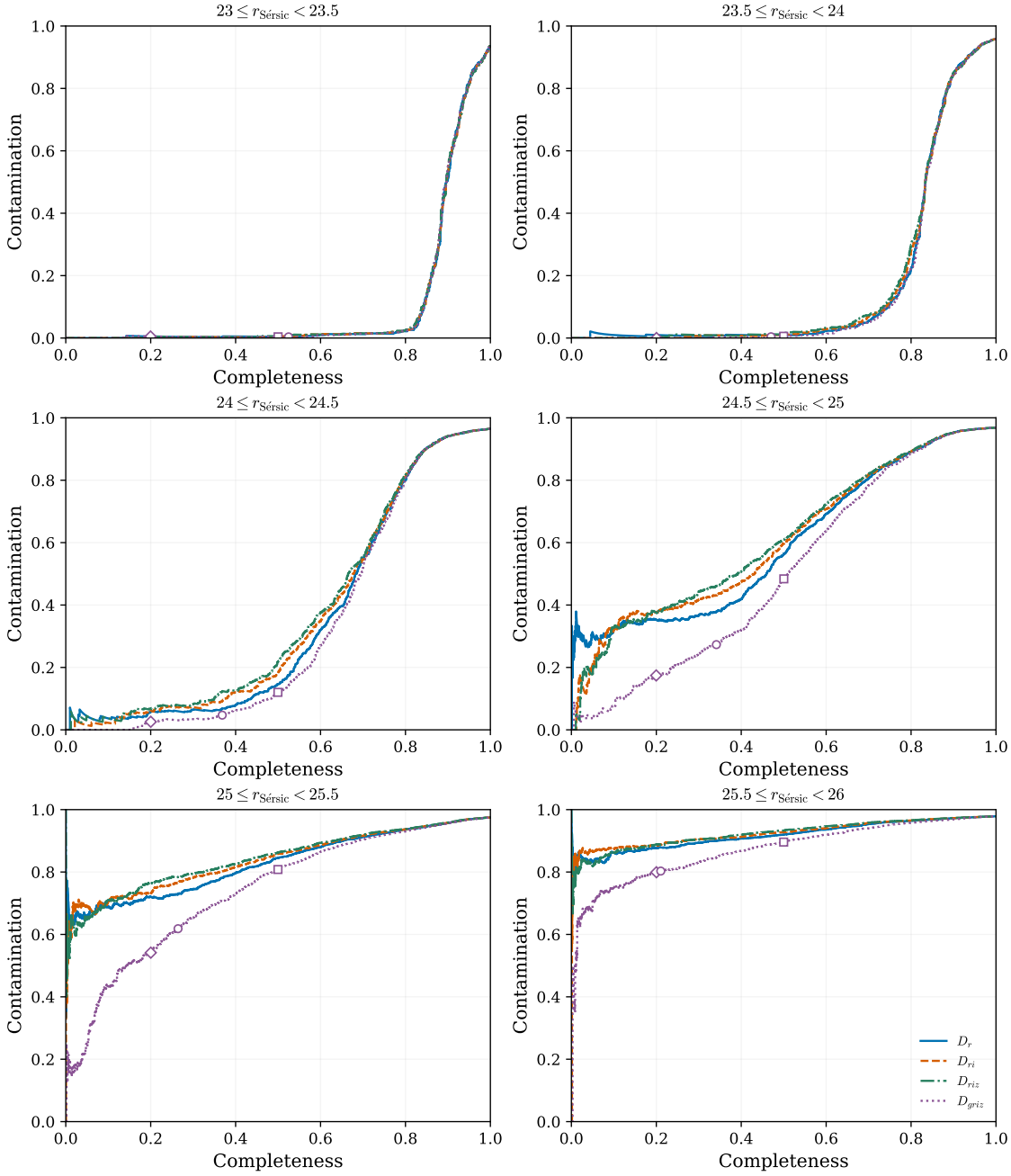


Figure 19. The performance of multi-band D classificators for stellar subsamples in different magnitude bins, as marked. Empirical contamination–completeness curves based on objects with HST classification labels are shown for the following band combinations: r, ri, riz and $griz$. All curves within a panel use the same cleaned HST reference sample with valid D in all required bands. Diamonds and squares mark approximately 20% and 50% stellar completeness on the $griz$ curves; circles mark $D_{griz} = 3$ threshold. Objects that are beyond the plotting window in Fig. 16 remain in the completeness and contamination denominators. Note that morphological information from additional bands improves performance.

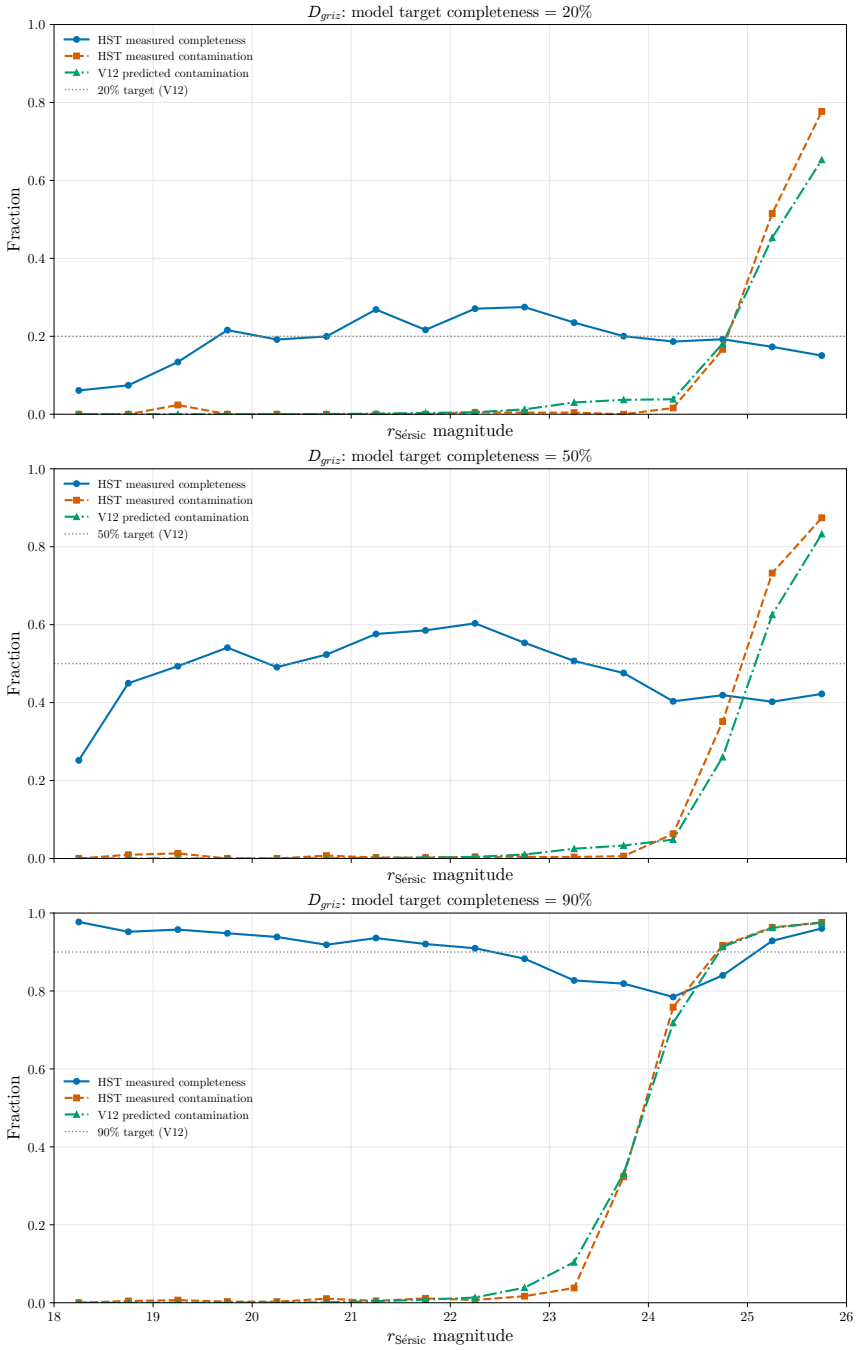


Figure 20. Target completeness application. We will fix caption after this figure is finalized.

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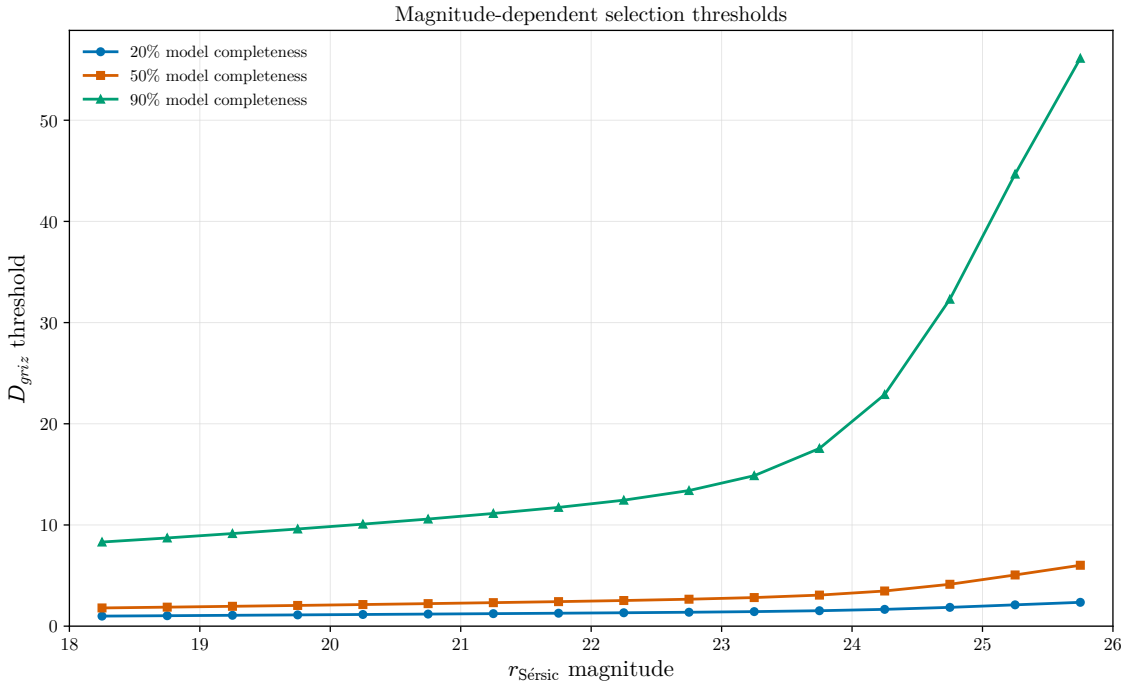


Figure 21. D values for target completeness application. We will fix caption after this figure is finalized.

Sesar, B., Ivezić, Ž., Lupton, R. H., et al. 2007, AJ, 134, 2236, doi: 10.1086/521819

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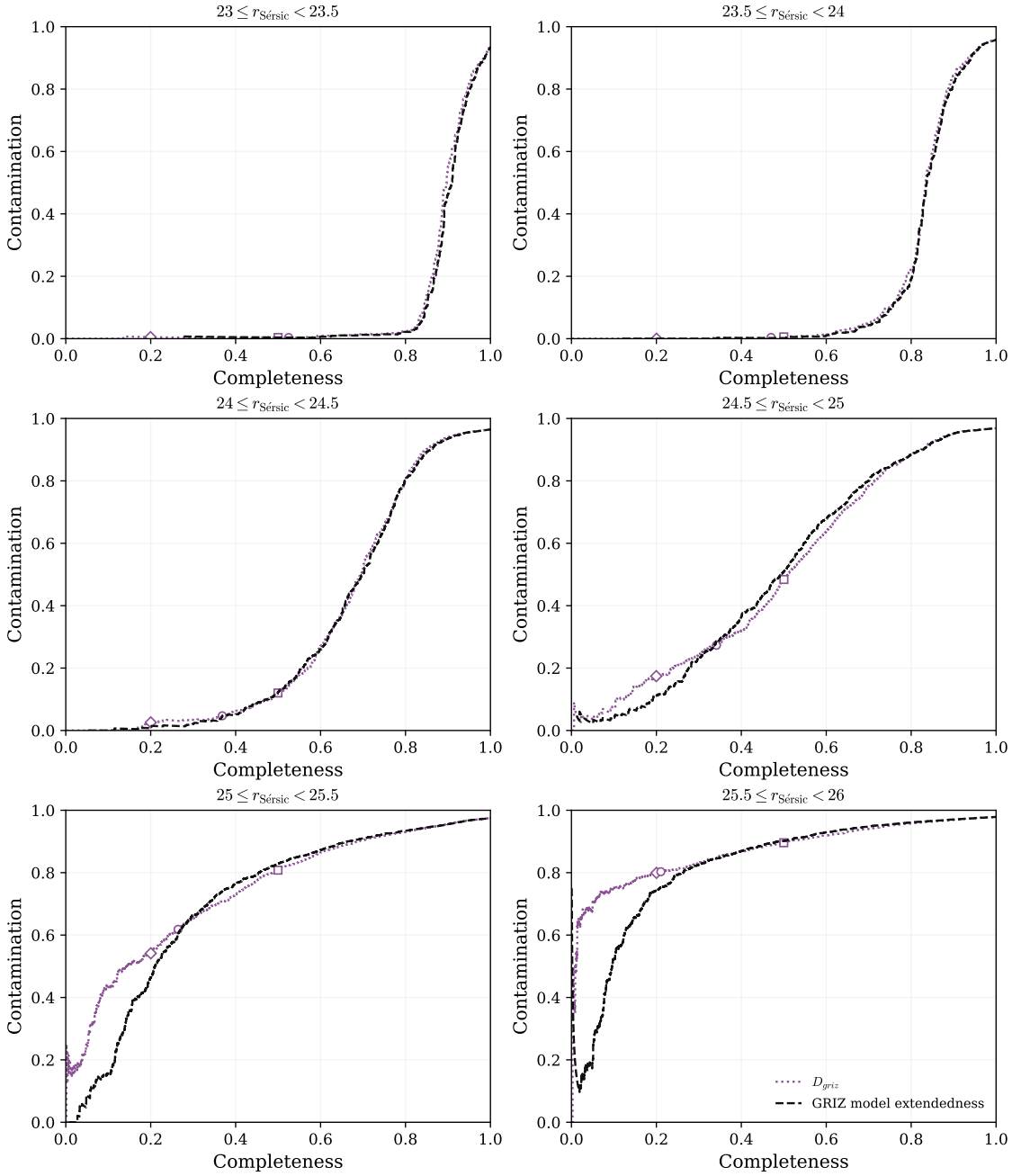


Figure 22. A comparison of completeness–contamination curves for two different classification methods based on D_{griz} developed here and `griz_model_extendedness` computed by Rubin Science Pipelines.

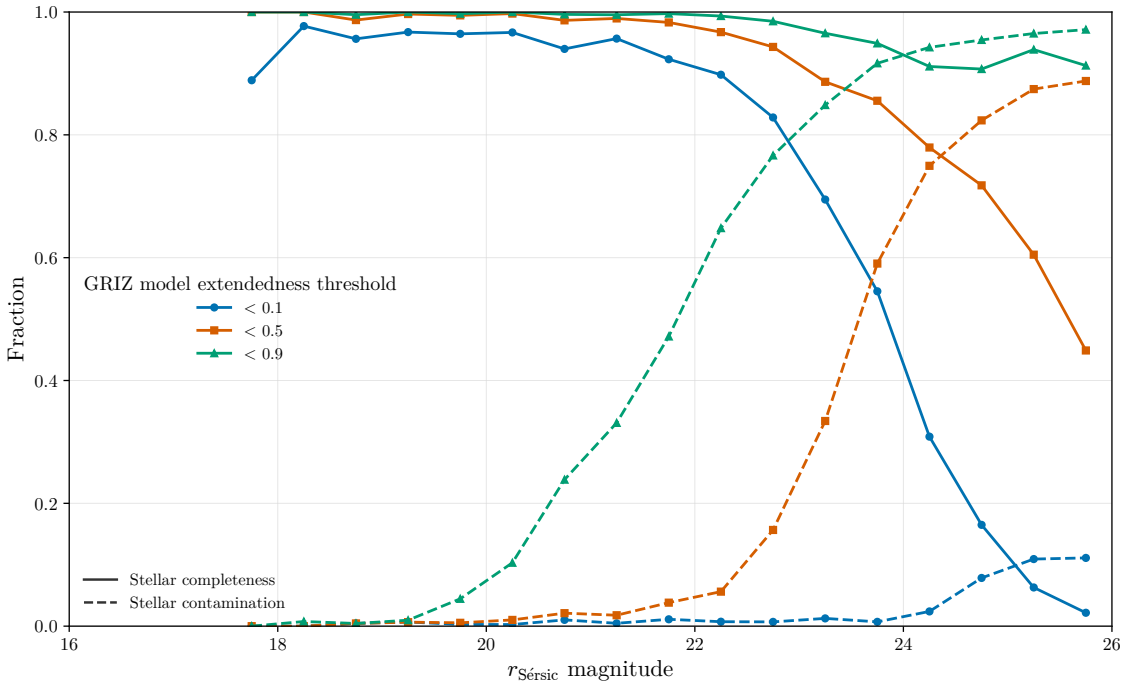


Figure 23. Performance vs. magnitude for a fixed threshold. IF we keep this figure, we will fix caption after this figure is finalized.

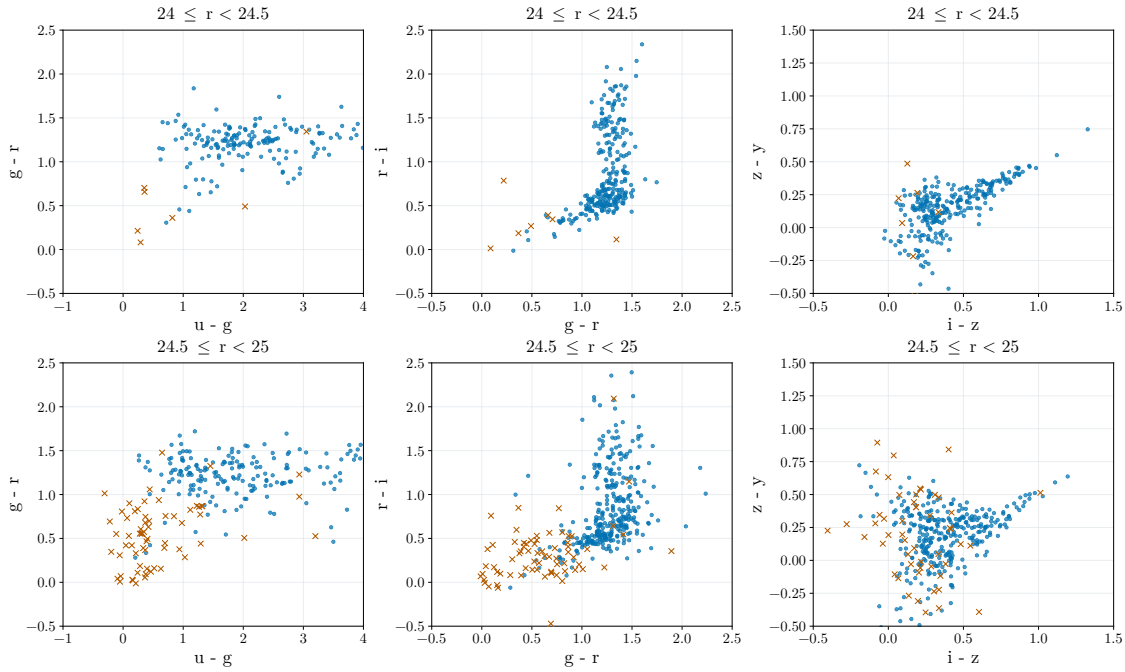


Figure 24. The color distribution of objects morphologically classified as unresolved using Rubin DP2 measurements (with D_{griz} thresholds selected to ensure 20% completeness), and then separated into stars (blue dots) and galaxies (red crosses) using HST labels, in two magnitude bins: $24 \leq r < 24.5$ (top) and $24.5 \leq r < 25$ (bottom). The fraction of HST-classified galaxies in these two subsamples is 2.6% and 18%, respectively. Note that galaxies appear systematically bluer than stars, except for the $z - y$ color, which allows for improved color-based classification.

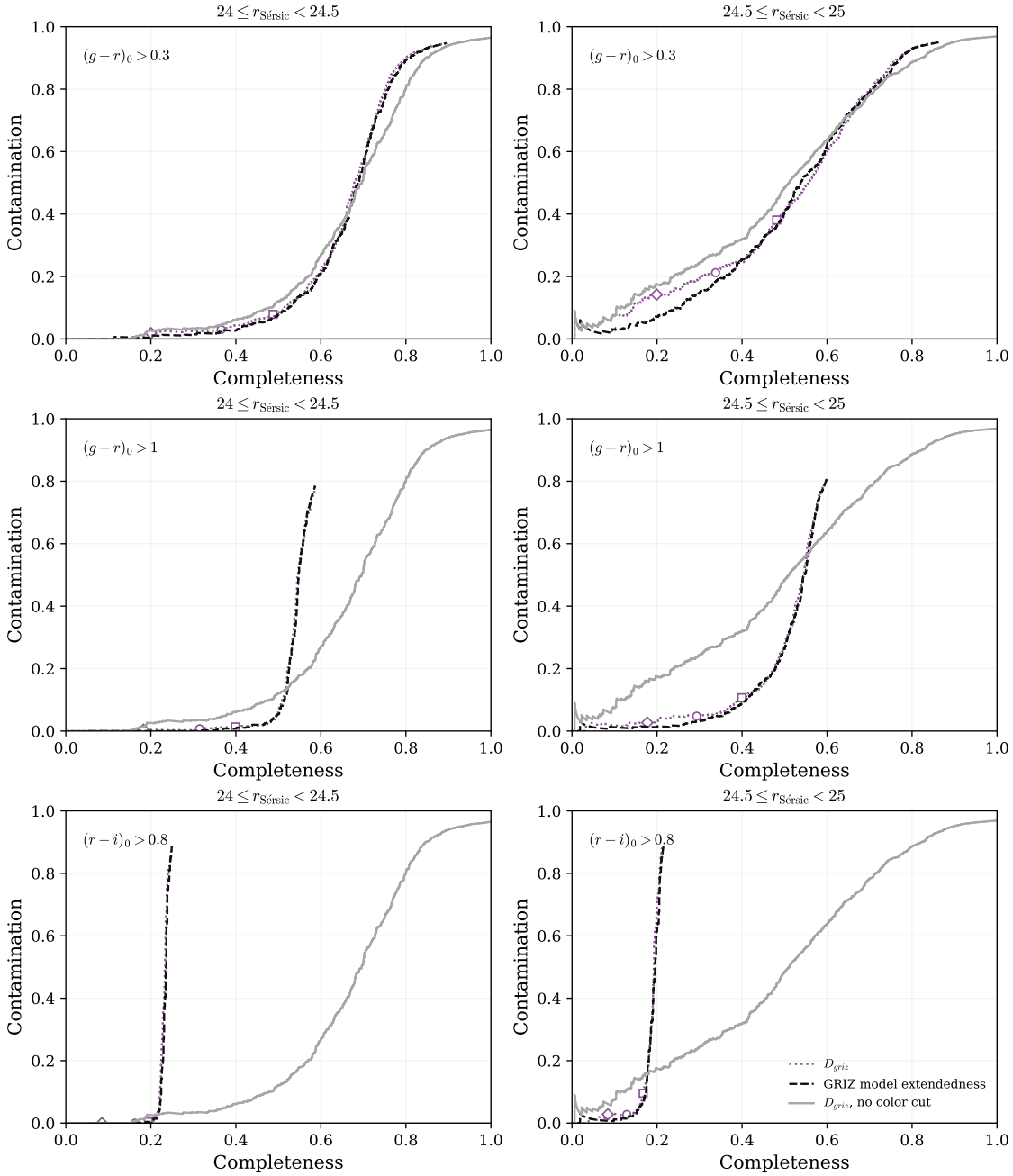


Figure 25. The performance of color-aided multi-band D classificators for stellar subsamples in two magnitude bins: $24 \leq r < 24.5$ (left) and $24.5 \leq r < 25$ (right), and for three color cuts: $g-r > 0.3$ (top), $g-r > 1.0$ (middle) and $r-i > 0.8$ (bottom). In each panel, empirical contamination-completeness curves based on objects with HST classification labels are shown for the following band combinations: r, ri, riz and $griz$. A curve corresponding to the $griz$ case without color cut is also shown for reference. Note the marked decrease of sample contamination when morphological classification is aided with color selection. Diamonds and squares mark the $griz$ thresholds corresponding to approximately 20% and 50% completeness before the color cut, respectively; circles mark $D_{griz} < 3$. Open gray diamonds show the 20% completeness selection without a color cut.